

How HFT Speed Disruptions Affect Market Quality in Energy ETF Markets

Abstract

We examine how high-frequency traders' (HFTs) speed advantages affect market quality in energy exchange traded-funds (ETFs) using weather-induced microwave disruptions. We show that the effects of latency arbitrage are heterogeneous across ETFs. While HFT effects were limited during initial technology adoption (2011–2012), they became substantial after Quincey Data expanded access in 2013. Disruptions to HFT speed increase trading activity across all energy ETFs. In less liquid markets, speed advantages widen spreads, increase price impacts, and raise volatility. In highly liquid markets price effects are negligible. For highly liquid ETFs, trades become more informative when speed advantages are neutralized, suggesting that the fastest traders introduce noise rather than fundamental information. These findings indicate that trading speed frictions can affect information quality and liquidity in energy financial markets.

Keywords: high-frequency trading; exchange-traded funds; market quality; latency arbitrage; price discovery, market microstructure

JEL Codes: G1, G2, Q4

1 Introduction

Energy futures exchange-traded funds (ETFs), representing over \$1.8 billion in assets under management,¹ provide retail and institutional investors exposure to energy futures via an underlying portfolio of futures contracts without the capital requirements and operational complexity of futures markets. Market quality in these ETFs affects trading conditions and the efficiency with which energy price information is incorporated into financial markets. Wider quoted spreads, which capture the costs of trading, and larger price impacts raise transaction costs and reflect greater adverse selection risk, while impaired price efficiency distorts risk management and asset-allocation decisions. Because energy ETFs are closely linked to underlying futures markets, frictions in their trading can affect how information about energy prices is transmitted and reflected across markets.

High-frequency traders (HFTs) play a central role in ETF markets. Comerton-Forde and Marta (2025) show that HFTs supply most liquidity in French equity ETFs and tighten spreads. Jain et al. (2021) find that HFTs reduce the magnitude and persistence of deviations between U.S. equity ETFs and the Net Asset Value (NAV) of the underlying portfolios. In a review covering equity, fixed income, and leveraged fund ETFs, Ben-David et al. (2017) highlight algorithmic traders and high-frequency arbitrageurs as important agents in the arbitrage between ETFs and their underlying assets. Budish et al. (2015) document that HFTs dominate ultra-short-horizon ETF-futures arbitrage in the S&P 500, while Dao et al. (2017) show that HFT arbitrage based on relative value between gold ETFs yields substantial returns. Despite this extensive research on HFTs in equity and bond ETFs, the role of HFTs in energy futures ETFs remains unexplored.

We study how latency arbitrage affects market quality in energy ETFs and how these effects vary across trading environments in markets linked to energy price formation. We focus on two highly traded ETFs, USO and UNG, providing exposure to crude oil and natural gas futures, respectively, with mean daily dollar volumes of approximately \$318.4 million

¹Based on 2025 SEC Form 10-K filings for USO, UNG, and UCO.

and \$118.3 million (2011–2025 Bloomberg). We contrast these with two less liquid crude oil products, UCO and OIL, which average roughly \$90.1 million (2011–2025 Bloomberg) and \$20 million (2011–2020 Bloomberg) per day. We group ETFs by trading activity, while recognizing that these products also differ in institutional features such as leverage, underlying commodity exposure, and structure.

In modern electronic markets, “being informed means seeing and acting on prices sooner than competitors” (Haldane, 2011). HFTs invest in technologies that minimize latency and enable rapid responses to new information. These investments are essential for exploiting fleeting price discrepancies across markets. Because public intraday market data do not identify trader accounts, we rely on latency variation induced by telecommunication infrastructure to identify HFT activity. Firms have built faster communication links between Chicago and New York exchanges—connecting the CME, where energy futures trade, with the New York exchange cluster (e.g., NYSE, Nasdaq, or Cboe), where energy ETFs trade. A high-speed fiber-optic cable was completed in 2010, and by late 2010 firms were seeking Federal Communications Commission (FCC) approval to construct microwave transmission routes. Microwave networks reduce transmission times from Chicago to New York by about 2 milliseconds relative to fiber optics (Laughlin et al., 2014). By September 2012, fifteen microwave networks had registered between Chicago and New York. In December 2012, Quincy Data began selling an ultra-low-latency market-data feed delivered over its microwave network, broadening access to faster Chicago–New York price information without requiring firms to build their own infrastructure. However, broader access did not eliminate latency hierarchies, as firms continued to differ in routing paths, hardware, and network optimization. Our analysis therefore focuses on HFTs operating at the speed frontier—those able to exploit the fastest available Chicago–New York connections. The sample is split into pre-Quincy (2011–2012) and post-Quincy (2013–2017) periods to capture how HFT impacts differ before and after this expansion in ultra-low-latency access.

Although microwave networks have improved in speed and capacity over time, they re-

main inferior to fiber optics in both bandwidth and reliability. Unlike fiber optics, microwave transmissions are disrupted by rain and snow, temporarily eliminating their latency advantage over fiber-optic routes. This provides a natural experiment that we use as our identification strategy. To determine when weather conditions meaningfully degrade microwave performance, we rely on International Telecommunications Union (ITU) rainfall-attenuation thresholds, and we map precipitation along the Chicago–New York corridor using Thiessen (Voronoi) polygons to spatially aggregate weather-station data. Our approach yields more conservative and physically grounded estimates of disruption than those produced by previously used data-driven methods (Shkilko and Sokolov, 2020). Consistent with a switch from microwave to fiber-optic routing, we document that NY-cluster energy ETFs react to CME price changes with a 2-millisecond delay during precipitation disruptions.

While this identification strategy isolates exogenous variation in trading speed, its implications for market quality depend on an economic mechanism linking speed to cross-market arbitrage and liquidity provision. Speed facilitates rapid arbitrage and quote adjustment between futures and ETFs, and impairing that speed slows adjustment and increases the risk of stale quotes. As a result, liquidity providers face greater adverse selection risk, which can lead to wider spreads, reduced depth, and changes in trading activity. These effects are unlikely to be uniform across ETFs. In less liquid ETFs, where price alignment with futures depends more on fast arbitrage, delays in information transmission can have larger effects on market quality. In more liquid and deeper ETFs, by contrast, the same shock may generate smaller effects because greater depth and liquidity provision help stabilize quotes. This yields a clear empirical prediction: rain-induced microwave disruptions should have larger market-quality effects in ETFs whose trading environment is more sensitive to fast arbitrage.

Using Putniņš (2013) information leadership shares, we document that energy price information flows from CME futures to NY-cluster energy ETFs, justifying our focus on HFT impacts in the ETF market. We find limited spatial arbitrage effects before 2013 but sub-

stantial and heterogeneous effects during the post-Quincy period using panel regressions of market quality on precipitation disruptions, while controlling for confounding market conditions. We document that HFT speed advantages reduce trading activity in both highly liquid and less liquid energy ETFs. In less liquid markets, these advantages additionally widen spreads, increase price impacts, and raise volatility, while highly liquid markets show no such price effects. Hasbrouck (1991) Structural Vector Autoregression (SVAR) estimates show that speed advantages reduce the information content of trades especially in the most liquid markets, suggesting the fastest traders introduce noise rather than fundamental information into price discovery.

Our methodological approach highlights the importance of threshold selection in precipitation-based identification strategies. Using thresholds grounded in microwave attenuation yields a more selective set of disruption intervals than data-driven approaches, and we show that this choice materially affects the estimated impact of latency disruptions on market quality.

This study makes three main contributions. First, we show that the market-quality effects of latency arbitrage are not uniform across assets, but depend on the trading environment in which energy-related prices are formed and transmitted across markets. In energy ETFs, disruptions to intermarket transmission have larger effects in less liquid products, while more liquid ETFs are better able to absorb short-run latency frictions. This heterogeneity arises because speed facilitates cross-market arbitrage and quote adjustment, and impairments to that speed increase adverse selection risk for liquidity providers, particularly in thinner markets. Second, we provide evidence that speed hierarchies persist even after the expansion of low-latency access through Quincy Data. While prior work suggests that broader access may compress speed differences, our results indicate that economically meaningful latency advantages remain, helping explain why the effects of latency arbitrage continue to shape market quality across ETFs. Third, we develop a precipitation-based identification strategy grounded in microwave signal attenuation and Thiessen-polygon aggregation, which links weather conditions directly to disruptions in low-latency communication networks and

provides a physically motivated measure of latency shocks.

2 Data and Empirical Methods

We obtain precipitation and market data from four data sources for the period from 2011 to 2017: the National Oceanic and Atmospheric Administration (NOAA), the NYSE Daily Trade and Quote (DTAQ) data, the CME Time and Sales data, and Bloomberg. From NOAA we obtain precipitation data to identify periods when precipitation may have disrupted microwave transmission. The U.S. 15-Minute Precipitation Data, archived by NOAA’s National Centers for Environmental Information, comprises measurements from approximately 2,000 weather stations across the United States. The dataset includes precipitation rates from Fisher & Porter gauges, which record precipitation in increments as small as one-hundredth of an inch and are timestamped at 15-minute intervals (section 2.2.).

The NYSE DTAQ reports intraday trade and quote data for all market activity within the U.S. National Market System. It provides tick-by-tick data capturing every individual trade and every change in posted bid and ask prices, timestamped to the millisecond for each market and exchange. This includes the National Best Bid and Offer (NBBO), which represents the highest bid and lowest ask prices across all exchanges, and consolidated trades. We use the NBBO and consolidated trades to assess market quality and trade informativeness for our sample of ETFs (USO, UNG, UCO, and OIL). Tables A.1 and A.2 in Appendix A illustrate the structure of these data.

From the CME, we obtain crude oil and natural gas futures Time and Sales market data (Table A.3 illustrates the data structure) timestamped to the second, which provides the price, quantity, and time of each executed trade. We use these data for cross-market price discovery analysis to verify that information flows from Chicago to NY. We additionally obtain CME Market Depth data, timestamped to the millisecond, for select months to confirm HFT spatial arbitrage between CME and the NY cluster. While the Market Depth data

allows reconstruction of the limit order book up to ten depths, we only use top-of-the book information (see Figure A.1 in appendix A). Daily closing prices and traded dollar volumes are sourced from Bloomberg (section 2.3.) to describe the trading activity for the ETFs studied.

Our analysis focuses on energy ETFs tracking crude oil (USO, UCO, OIL) and natural gas (UNG), paired with their underlying CME futures contracts (CL and NG). We organize the remainder of this section as follows: in the first subsection we discuss the direction of spatial latency arbitrage. Second, we describe precipitation measurement. Third, we show that the computed precipitation measurements serve as an identification strategy for the presence of HFTs that engage in spatial latency arbitrage between CME energy futures and NYSE ETF markets. Fourth, we discuss our empirical approach to study ETF market quality measures.

2.1 Information Flows

To determine which market’s quality is affected by spatial latency arbitrage, we first establish the direction of information flows between CME futures and NY-cluster ETFs. Market events provide valuable information that HFTs may use for spatial latency arbitrage. Following previous research (Shkilko and Sokolov 2020), we hypothesize that spatial arbitrage follows the direction of market information flows.

To establish the direction of cross-market information flow, we use the information leadership share (ILS) measure of Putniņš (2013). ILS is well suited to our setting because ETF and futures prices are cointegrated, while the two markets may differ in transitory noise and quote adjustment speeds. In such settings, conventional price-discovery measures can confound informational leadership with relative noise, whereas ILS isolates the permanent-information component. This allows us to verify the economically relevant direction of information transmission between CME futures and NY-cluster ETFs before turning to ETF-side market quality effects. As a robustness check, Appendix B reports Hasbrouck

information shares, which yield the same qualitative conclusion: price discovery is centered in CME futures.

Across all pairs, the leadership ranking is economically large and stable across measures, so the subsequent analysis focuses on how ETF market quality responds when the speed advantage of cross-market arbitrageurs is disrupted.

CL futures' ILSs exceed 75% when paired with oil ETFs (USO, UCO, OIL), and similar dominance exists for NG-UNG pairs. These results confirm that energy price information predominantly flows from CME futures to NY securities, justifying our focus on how spatial arbitrage affects market quality in the ETF market.

2.2 Measuring Precipitation

We use precipitation to identify HFTs trading at the frontier of speed because precipitation attenuates radio signals, disrupting microwave transmissions and eliminating their speed advantage over fiber optics. We obtain 15-minute precipitation data from NOAA weather stations along the Chicago-New York corridor.

Shkilko and Sokolov (2020) calculate arithmetic average precipitation across selected weather stations, defining heavy precipitation as intervals exceeding half or one standard deviation above quarterly means ($P(1/2\text{StD})$ and $P(1\text{StD})$). We adopt a more conservative identification approach by employing International Telecommunication Union (ITU) precipitation-attenuation thresholds, which are based on the physics of signal disruption rather than sample-based statistical cutoffs. We also use Thiessen (1911) polygons to weight each station by its geographic area of influence when aggregating precipitation spatially. We consider two precipitation measures: $P(\text{ITU})^*$ using our proposed approach, and $P(1\text{StD})$ for comparison with prior research. Figure 1 illustrates our partition of the Chicago-New York corridor into 75 polygons, each affiliated with a weather station. Appendix C offers methodological details.

Our objective is not to identify intervals with unusually high rainfall per se, but inter-

vals in which precipitation is likely to impair microwave transmission along the Chicago–New York corridor. For this reason, the main treatment variable is based on ITU attenuation guidance combined with corridor-level Thiessen aggregation, rather than on a sample-dependent rainfall cutoff. An endogenous threshold such as one standard deviation above mean precipitation captures unusually high rainfall relative to the sample, but does not align directly with the transmission mechanism of interest. In this setting, that distinction matters because rainfall that does not materially affect microwave attenuation can still enter the treatment set, introducing measurement error.

Table 1 compares precipitation measures across methods. $P(\text{ITU})^*$ averages 0.00154, with 3.5% of intervals exceeding ITU thresholds. In contrast, $P(1 \text{ StD})$ averages 0.00241, with 10% of intervals exceeding the 1 StD threshold. These differences are statistically significant (Wilcoxon signed-rank test $p < 0.01$), suggesting threshold selection may affect economic conclusions. In the next subsection we investigate how precipitation disrupts microwave transmission and serves as an identification strategy for HFT spatial arbitrage between Chicago and NY markets.

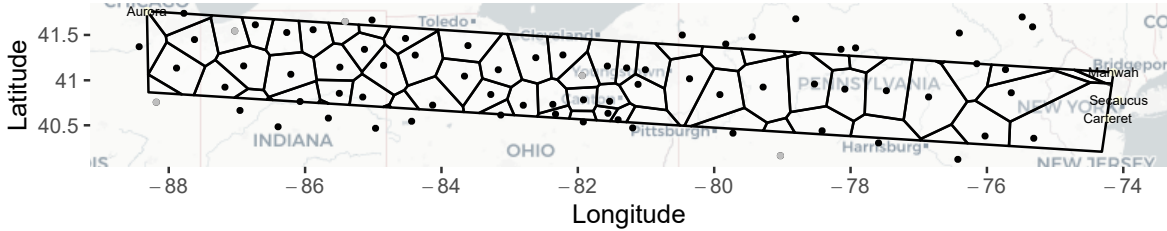


Fig. 1. Thiessen Polygons Inscribed on a Spherical Quadrangle

Note: This figure illustrates the partition of the area covering the microwave network corridor between Chicago and New York. The northwest and northeast vertices are located on the data centers of the CME in Aurora, Illinois and the NYSE in Mahwah, New Jersey. The spherical quadrangle includes Carteret and Secaucus, New Jersey since these locations house the data centers for the Nasdaq and the Cboe. Based on Laughlin et. al., (2014), we use a spherical quadrangle to encompass most microwave transmission networks. The polygon vertices are generated using US National Atlas Equal Area (EPSG:2163) and forced on World Geodetic System (EPSG:4326). We partition the quadrangle using Thiessen polygons, each affiliated with the weather station representing the corresponding area. Precipitation is calculated as a weighted average where each station’s value is weighted by the area it

represents. The black and gray points represent the locations of the weather stations used to generate the Thiessen polygons. Gray points represent weather stations that close at some point during the period of study. Further details are provided in appendix C.

2.3 Identification Strategy

Our identification strategy relies on precipitation disrupting microwave networks, which slows down the fastest HFTs. Microwave transmissions typically require 4.5-5.5 milliseconds between Chicago and New York, while fiber optic transmissions require 6.5-8 milliseconds, creating a 2-3 millisecond difference (Laughlin et al. 2014). We validate this identification strategy by examining whether ETF trading patterns shift during precipitation disruptions in ways consistent with a switch from microwave to fiber-optic routing.

We consider futures trades as signals that HFTs may respond to by trading in the NY cluster ETFs. Similar to Rosch (2021), we define a selected futures price signal using two distinct criteria: the occurrence of a trade not preceded by another trade within a 10-millisecond time frame, or a trade that represents a change in price direction per the Lee and Ready (1991) algorithm. By requiring either time separation or direction changes, we aim to capture genuine HFT activity. We examine trade responses within 5-10 milliseconds of these signals.² We count security trades following futures signals during both precipitation-disrupted and non-disrupted intervals using selected months with sufficient precipitation events (see appendix D for details).

Figure 2 presents results for the CL-USO pair, our most liquid EFT pair where HFT activity is most likely to be observed, illustrating how precipitation affects HFT trading behavior. When there are no precipitation disruptions according to the Thiessen-ITU criteria, ETF trades occur 5-6 milliseconds after a signal. During disruptions, trades occur 7-8 milliseconds after a signal. These results strongly suggest a shift from microwave to fiber optics, confirming that precipitation effectively disrupts the speed advantages of the fastest HFTs. In the next subsection, we outline the measures used to assess market quality.

²Beyond 10ms, trades in the NY cluster are more likely to be related to other signals.

Table 1. Aggregate Precipitation Rate Statistics in Inches per 15-minute Interval

	Min	Mean	Median	St. Dev	Max	% of Int w. Precip
P(1 StD)	0	0.00241	0	0.00853	0.21622	10.49909
P(ITU)*	0	0.00154	0	0.00847	0.25458	3.78783

Note: This table presents summary statistics on aggregate precipitation and percent of intervals with precipitation (% of Int w. Precip) on the Chicago – New York corridor. Summary statistics are expressed in inches for each 15-minute interval during market trading hours (9:30 – 4:00 pm EST) for the period from 2011 to 2017. Total number of intervals is 47,547. We compare P(ITU)* our proposed method based on Thiessen rain aggregation and ITU precipitation thresholds, and P(1Std) based on average precipitation and endogenous precipitation thresholds. We test whether median precipitation measures differ using the Wilcoxon signed-rank test. Jarque-Bera tests for normality are rejected in all instances.

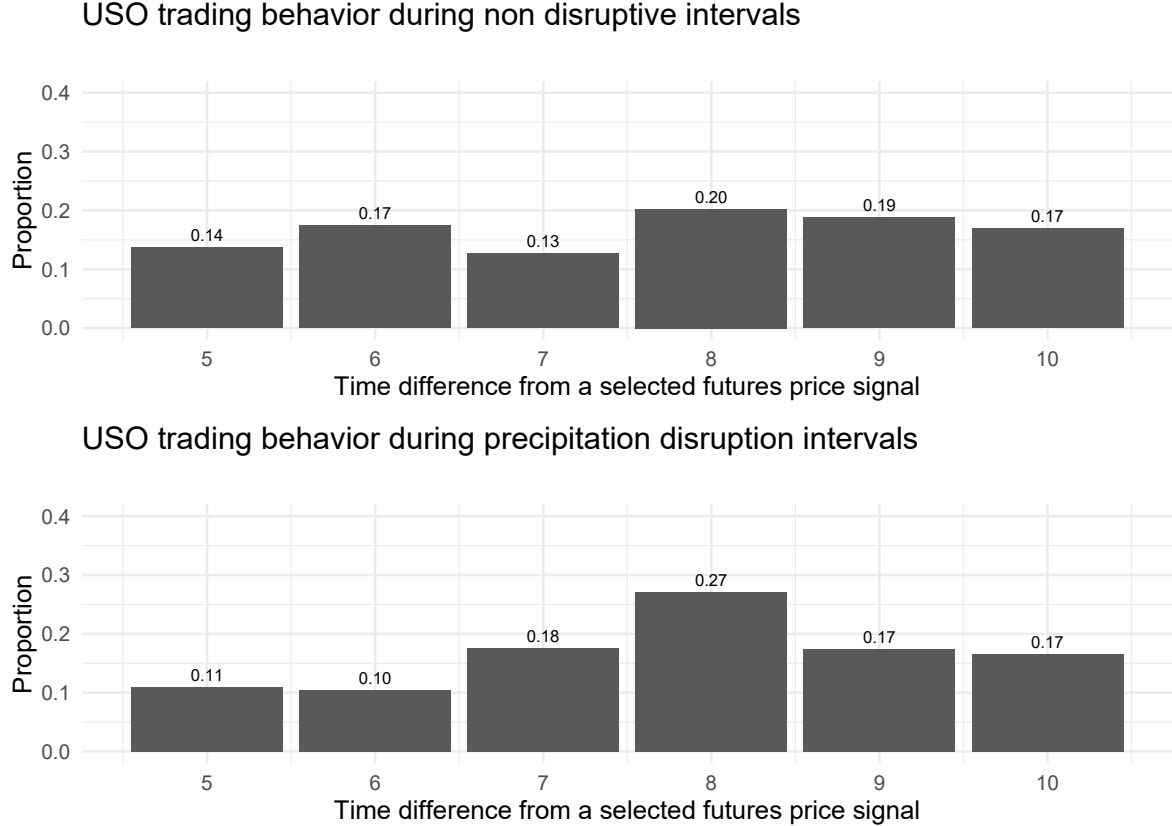


Fig. 2. USO Trades Following CL Futures Price Signals

Note: This figure presents the USO ETF reactions to selected futures price signals originating from the CL futures. We count the number of trades that occurred in the spatially separated exchange and bin the counts per rounded millisecond time difference from the selected futures price signals. The y-axis measures the proportions of trade counts. The months of June 2013 and March 2016 are selected due to a higher incidence of disruptive precipitation periods during Mondays, Thursdays, and Fridays which removes concerns about distortions created by the American Petroleum Institute and Energy Information Administration announcements. See appendix D for further details.

2.4 Market Quality Measures and Empirical Approach

To assess the impact of HFTs on the energy ETFs traded on the NY markets, where information from CME is arbitrated, we process the DTAQ data, time-stamped to the millisecond (t^{ms}), by merging trade and quote data matching each trade to the prevailing best bid and offer. Figure A.2 in Appendix A shows that most of NBBO updates derived from either trades or quotes across the NY cluster exchanges are driven by activity on NYSE, Nasdaq,

and Cboe. We compute market quality measures for each ETF at the second (t^s) level, and then summarize these measures at 1-minute interval (t). The 1-minute frequency balances two considerations: it is fine enough to preserve intraday variation in market quality, yet coarse enough to avoid excessive noise when matching with our 15-minute precipitation data (discussed below).

Table 2 presents the market quality measures and their definitions. We adopt the effective spread (ES) as our primary trading cost metric, defined as twice the signed difference between the prevailing midquote $m_{t^s} := (NBBO\ Ask_{t^s} + NBBO\ Bid_{t^s})/2$ and the trade price p_{t^s} . Formally, $ES_{t^s} = 2q_{t^s}(p_{t^s} - m_{t^s})$, where q_{t^s} indicates trade direction ($q_{t^s} = 1$ for buy-initiated trades and $q_{t^s} = -1$ for sell-initiated trades). We utilize the Lee and Ready (1991) trade direction algorithm to obtain q_{t^s} , as this information is not available in the DTAQ data.

The realized spread (RS) represents the market making revenue and is defined as $RS_{t^s} = 2q_{t^s}(p_{t^s} - m_{t^s + \delta})$. The price impact (PI) represents the adverse selection risk, measured as the change in the security's value following a trade, defined as $PI_{t^s} = 2q_{t^s}(m_{t^s + \delta} - m_{t^s})$. We define δ as 5 seconds ($\delta = 5$), the smallest interval recommended by the literature (O'Hara 2015). ES , RS and PI are linked through the following equation $RS_{t^s} = ES_{t^s} - PI_{t^s}$ which shows that the revenue generated by liquidity providers is the effective spread adjusted for price impacts. To compute ES , RS , and PI for each minute t , we first derive their components using event-time, time-stamped to the millisecond (i.e., $p_{t^{ms}}$, $m_{t^{ms}}$, $q_{t^{ms}}$). Using the previous-tick method, we then measure the components at the second-level (p_{t^s} , m_{t^s} , q_{t^s}) and derive ES_{t^s} , RS_{t^s} , and PI_{t^s} . We finally take the median values within each 1-minute interval to produce ES , RS , and PI . Following Shkilko and Sokolov, we report ES s, RS s, and PI s in raw price units rather than converting them to basis points. This maintains comparability with the existing empirical design and preserves a direct interpretation of these measures as trading costs.

Table 2. Market Quality Measures and their Descriptions

Market Variable	Brief Description	Units
Effective Spread (ES)	Trading cost equal to twice the signed difference between the prevailing midquote m_t and the trade price p_t . $ES_t = 2q_t(p_t - m_t)$, where q_t is the trade direction indicator.	USD
Realized Spread (RS)	Market making revenue. $RS_t = 2q_t(p_t - m_{t+\delta})$, where $m_{t+\delta}$ is the forward midquote price.	USD
Price Impact (PI)	Adverse selection risk. $PI_t = 2q_t(m_{t+\delta} - m_t)$.	USD
Realized Volatility (RV)	$RV = \sqrt{\sum r_t^2}$, where r_t are log midquote returns.	Unitless
Market Depth (MD)	Number of contracts at the top of the book on the bid and ask sides.	Count
Traded Volume (VOL)	Number of shares traded.	Count
Number of Trades (TS)	Number of individual trades.	Count
NBBO Quote Updates (NBBO)	Updates to the National Best Bid and Offer.	Count
Price Discovery Shares (IS)	Hasbrouck (1991) information share; computation details in Appendix B.	Unitless

Note: This table presents a brief definition of the market variables used in our analysis. The index t represents 1-minute intervals. ES, RS, and PI are the median values within each 1-minute interval, while RV, VOL, TS, Market Depth, and NBBO Quote Updates are the sums during 1-minute intervals. The Lee and Ready (1991) algorithm is used to determine the trade direction indicator q_t .

Beyond these trading cost measures, we also examine market dynamics and trading activity. Realized volatility (RV) quantifies return variation over time. To compute RV , we first calculate the returns of the natural logarithm of midquote using event-time, $r_{tms} = \ln(m_{tms}) - \ln(m_{tms-1})$. By using the previous-tick method, we derive the second-level midquote returns, r_{ts} . Finally, we derive RV_t by summing the squared returns within each minute: $RV_t = \sqrt{\sum_j^I r_j^2}$, where I is the number of second-level events within minute t , and r_j represents the midquote return for the j -th second-level event. While the previous-tick method for ES, RS, PI, and RV might result in a loss of intra-second information, this should not affect the results of our statistical procedures that rely on 1-minute data.

We measure trading activities through the traded volume (VOL), defined as the number of shares traded per minute, and the number of trades (TS), measured as the count of individual trades per minute. We further study the total number of NBBO quote updates

per minute (NBBO) contributed by each market. For these measures, we aggregate data from event time to the minute level using summation. This approach preserves the cumulative information from the millisecond level, ensuring that the full extent of the data is retained in the 1-minute aggregations. We include market depth (MD) as another indicator of market quality, defined as the number of contracts at the top of the book (NBBO) on both the bid and the ask side per minute.

Having defined our market quality measures, we now describe our empirical strategy for identifying HFT effects. We assess the impact of HFTs using microwave technology on market quality by using 15-minute precipitation data as an exogenous intraday shock that disrupts microwave transmissions and narrows speed differences among traders. The difference in time granularity between the market and precipitation data raises the challenge of combining the minute-level market quality measures with the 15-minute precipitation data. Given the rapid pace of market operations, aggregating market quality to 15-minute intervals is likely to result in information loss and biased results. Therefore, we analyze 1-minute intervals, assuming constant precipitation within each interval, equal to 1/15th of the original 15-minute observation. After these transformations, both the precipitation and DTAQ data can be merged based on their common calendar time.

We use a panel data approach to assess the effects of HFTs on market quality across energy ETFs. We split the sample into two periods, before and after Quincy data. We also estimate separate regressions for the two ETF groups based on liquidity characteristics: highly traded ETFs (USO and UNG) and less liquid ETFs (UCO and OIL). This separation allows us to examine whether HFT impacts differ based on baseline market liquidity. The panel regression for each ETF group is defined as:

$$Y_{i,t} = \beta_1 BP(\cdot)_{i,t} + \beta_2 \ln(VOL_{i,t-1}) + \beta_3 RV_{i,t-1} + \alpha_i + \sum_{c=1}^3 \mu_{ct} + \varepsilon_{i,t} \quad (1)$$

where index t represents minute-by-minute observations and index i represents individual securities within each ETF group. The dependent variable $Y_{i,t}$ is a market quality indi-

cator (Table 2). We convert our precipitation measures ($P(\cdot)_{i,t}$) to binary variables, where $BP(\cdot)_{i,t} = 1$ when there is a precipitation disruption for security i at time t , and 0 otherwise. We opt for this binary representation because an increase in the precipitation rate during a disruption should not substantially change the response of $Y_{i,t}$. For comparison with previous research, we compute equation (1) for BP(ITU)*, and BP(1 St.D).

We control for ETF-specific characteristics by including individual fixed effects α_i , and for systematic intraday patterns using three time fixed effects μ_{ct} , $c = 1, 2, 3$: a market open fixed effect, absorbing variation within the first 20 minutes of trading (μ_{1t}); a market close (μ_{2t}) fixed effect, capturing variation within the last 5 minutes of trading; and a market day fixed effect (μ_{3t}), encompassing the period between the market open and close.³ To account for error term heteroskedasticity (autocorrelation) and cross-sectional dependence in our panel setting we implement Driscoll-Kraay standard errors $\varepsilon_{i,t}$ (Driscoll and Kraay 1998).

Following the finance literature, we use RV and VOL as controls for market quality measures (Brogaard et al., 2015; Shkilko and Sokolov, 2020) in our panel regression specification. When estimating equation (1) with $VOL_{i,t}$ and $RV_{i,t}$ as dependent variables, we exclude them from the right side of the equation. To address potential endogeneity issues arising from using $VOL_{i,t}$ and $RV_{i,t}$ as control variables, we estimate the panel regression using their first lag.

To determine whether spatial latency arbitrage activities alter the information content of security trades, we employ a within-ETF price discovery approach based on the SVAR methodology of Hasbrouck (1991).⁴ The approach allows us to estimate how trades contribute to incorporating information into the efficient market price. This contribution is measured as information shares (IS), as detailed in appendix B. Price discovery is a continuous process that varies with public and private information, often differing across days as

³These intervals were chosen by visually inspecting intraday market volatility. See appendix E Figure E.1 for details. Choosing smaller opening intervals, e.g., 10 minutes for market open, does not affect regression results.

⁴This contrasts with the cross-market price discovery analysis discussed in section 2.1.

this information changes. To analyze how this process shifts due to HFT speed advantages, we estimate the IS separately for three distinct periods relative to each identified precipitation interval: before, during and after.⁵ The before and after periods correspond to the 15 minutes immediately preceding and following a precipitation disruption interval that do not register precipitation. The during period covers the minutes corresponding to the precipitation disruption interval. For each of these periods, we estimate a separate IS. To isolate the effect of precipitation, we compute a combined no-disruption measure by averaging the information shares from the before and after periods and compare this with the IS during disruptions using non-parametric Wilcoxon tests.

3 Empirical Analysis and Discussion

This section presents the results of assessing how disrupting HFT speed advantages affects market quality. Table 3 provides summary statistics for market quality measures during precipitation disruptions versus non-disruption periods, derived using the P(ITU)* metric and defined as in section 2.4 for our sample of energy ETFs. Panels A and B show summary statistics for the 2011-2012 and 2013-2017 periods with separate estimates for highly liquid (USO-UNG) and less liquid (UCO-OIL) ETFs. Jarque-Bera tests indicate non-normal distributions for all market quality measures.

Across both periods and precipitation conditions, effective spread (ES) and realized spread (RS) means differ by liquidity group. For highly liquid ETFs, ES and RS means range from \$0.0093 to \$0.0099, closely clustered around the \$0.01 tick size. For less liquid ETFs, ES and RS means range from \$0.0118 to \$0.0241, approximately 1.2 to 2.4 times the tick size. These patterns indicate that trading costs in highly liquid ETFs are near the minimum possible values, while less liquid ETFs exhibit wider spreads reflecting lower competition among liquidity providers. Price impact (PI) also varies systematically across liquidity groups. For

⁵There are very few instances where the precipitation intervals coincide with the market open. These are not excluded from the analysis.

highly liquid ETFs, PI means range from \$0.0002 to \$0.0005 across both periods, indicating minimal price impacts from trades. For less liquid ETFs, PI means range from \$0.0012 to \$0.0054, approximately 5 to 10 times larger than in liquid ETFs. The substantially higher price impacts in UCO-OIL reflect the greater difficulty of executing trades without moving prices in markets with limited depth. The relationship $RS_{ts} = ES_{ts} - PI_{ts}$ reveals different liquidity provision dynamics across market types. For highly liquid ETFs, liquidity provider revenues (RS) closely approximate the effective spread (ES), with price impacts capturing only 2-5% of the spread. This indicates that liquidity providers retain most of the spread as revenue, facing minimal adverse selection costs. For less liquid ETFs, price impacts represent a larger share of the effective spread—ranging from 8% to 23%—indicating that liquidity providers face greater adverse selection risk and capture a smaller portion of the spread as profit.

Comparing precipitation disruption periods to non-disruption periods reveals that market activity consistently increases during disruptions while spread and price impact measures either remain unchanged or decline. In the post-Quincy period (Panel B), USO-UNG volume increases from 30,264 to 33,567 shares, trade counts rise from 43 to 46 per minute, and market depth increases from approximately 240 to 260 thousand shares on both bid and ask sides. UCO-OIL shows similar patterns with volume rising from 9,075 to 9,901 shares and trade counts from 19 to 20. Higher market activity generates more NBBO updates across both groups. These raw means suggest improved market quality during precipitation disruptions, though formal testing is required to control for confounding factors.

The large standard deviations across all measures, which often exceed means indicate substantial variation within each category, making simple comparisons of means insufficient to discern the effects of precipitation disruptions. We therefore employ panel regression methods that control for lagged volume and volatility, as well as ETF-specific and time fixed effects, to isolate the impact of precipitation disruptions on market quality.

Table 3. Summary Statistics of Market Quality Indicators for the Sample of Energy ETFs

	Mean All	Mean P	Mean nP	St Dev All	St Dev P	St Dev nP
Panel A: 2011–2012 USO–UNG						
ES (USD)	0.0099	0.0098	0.0099	0.0063	0.0024	0.0064
RS 5s (USD)	0.0096	0.0095	0.0097	0.0075	0.0046	0.0076
PI 5s (USD)	0.0005	0.0005	0.0004	0.0040	0.0036	0.0040
RV (Unitless)	0.0007	0.0008	0.0007	0.0098	0.0097	0.0098
Bid MD (Count)	190515	200606	190028	250533	249912	250553
Ask MD (Count)	190511	200604	190024	250532	249915	250553
VOL (Count)	25380	26228	25339	58996	57218	59080
TS (Count)	54	60	54	103	99	103
NBBO (Count)	664	704	662	729	753	728
Observations	151495	6974	144521			
Panel A: 2011–2012 UCO–OIL						
ES (USD)	0.0240	0.0212	0.0241	0.1585	0.0295	0.1621
RS 5s (USD)	0.0202	0.0198	0.0202	0.0613	0.0222	0.0626
PI 5s (USD)	0.0054	0.0051	0.0054	0.1499	0.1582	0.1495
RV (Unitless)	0.0033	0.0031	0.0033	0.0176	0.0189	0.0175
Bid MD (Count)	14318	14762	14297	23816	25266	23743
Ask MD (Count)	14311	14757	14289	23834	25276	23762
VOL (Count)	3332	3679	3315	21085	10025	21475
TS (Count)	10	11	10	22	22	22
NBBO (Count)	433	434	433	532	510	534
Observations	151365	6973	144392			
Panel B: 2013–2017 USO–UNG						
ES (USD)	0.0096	0.0096	0.0096	0.0104	0.0051	0.0107
RS 5s (USD)	0.0093	0.0093	0.0093	0.0096	0.0057	0.0099
PI 5s (USD)	0.0002	0.0002	0.0002	0.0023	0.0023	0.0023
RV (Unitless)	0.0003	0.0003	0.0003	0.0057	0.0015	0.0059
Bid MD (Count)	241389	259144	239944	381430	433175	376868
Ask MD (Count)	241388	259144	239943	381430	433176	376868
VOL (Count)	30513	33567	30264	77206	78151	77123
TS (Count)	43	46	42	81	91	80
NBBO (Count)	311	321	311	400	411	399
Observations	539772	40636	499136			
Panel B: 2013–2017 UCO–OIL						
ES (USD)	0.0128	0.0125	0.0129	0.0363	0.0141	0.0376
RS 5s (USD)	0.0120	0.0118	0.0120	0.0320	0.0140	0.0330
PI 5s (USD)	0.0012	0.0010	0.0012	0.0168	0.0084	0.0173
RV (Unitless)	0.0007	0.0006	0.0007	0.0047	0.0029	0.0049
Bid MD (Count)	50059	49321	50119	88386	84189	88718
Ask MD (Count)	50035	49300	50095	88393	84195	88725
VOL (Count)	9075	9901	9008	26871	28860	26702
TS (Count)	19	20	19	43	51	42
NBBO (Count)	262	268	262	395	414	393
Observations	539423	40551	498872			

Note: This table presents summary statistics of market quality indicators for the sample of securities. We split the sample due to the Quincy Data structural break into 2011–2012 and 2013–2017. We report summary statistics separately for liquid and less-liquid ETFs. Statistics are derived from the millisecond DTAQ data and aggregated into 1-minute intervals. We report market quality measures discussed in section 2.4, a summary of these if found in Table 2. The P and nP indices to the column names refer to precipitation disruption and non-disruption sets respectively. Our precipitation disruption metric P for this table is BP(ITU)*.

3.1 Parametric Results

We assess the impacts of HFTs operating at frontier speeds on market quality using a panel regression (equation 1). We present results for the pre and post-Quincy periods in Table 4 using BP(ITU)* as the primary metric, with separate estimates for highly liquid ETFs (USO-UNG) and less liquid ETFs (UCO-OIL).

For the 2011-2012 period, results reveal insignificant coefficients at the 5% level for nearly all market quality measures in both ETF groups (Panel A, Table 4). The USO-UNG group shows a significant decline in RS by \$0.0001, representing 1% of the tick size, while the UCO-OIL group exhibits a significant \$0.0021 decline in ES, 21% of the tick size. These results indicate that HFT effects in the pre-Quincy period are concentrated in spread measures rather than broader market quality indicators, with stronger effects in less liquid ETFs. In these markets, HFTs appear to use their microwave-speed advantage to demand liquidity, raising trading costs; once this speed advantage is removed, their behavior shifts and costs fall. While RS declines during precipitation intervals in the USO-UNG markets, neither ES nor PI show any significant change, suggesting that HFTs do not materially alter trading or adverse selection costs in the most liquid markets. Results using BP(1 St.D) in appendix E (Panel A, Table E.1) yield a greater number of significant coefficients for non-spread measures (RV, MD, VOL).

Results using the broader endogenous rainfall cutoff are reported only as a comparison, rather than as an alternative baseline treatment definition. Because that rule is not tied to microwave attenuation, it admits a substantially larger set of rain intervals, many of which may not correspond to meaningful disruptions in microwave transmission. The resulting estimates should therefore be interpreted as reflecting treatment dilution and potential classification error under a meteorological, rather than technological, cutoff. For this reason, we do not further interpret these results.

The post-Quincy 2013-2017 period reveals heterogeneous effects of precipitation disruptions on market quality across ETF groups (Panel B, Table 4). For highly liquid ETFs,

results show coefficients for ES, RS, and PI near zero and statistically insignificant, suggesting that precipitation disruptions have minimal direct impact on prices in these markets. However, VOL and TS increase significantly by 12.8% and 2.893 units (approximately 6.9% of the average number of trades reported in table 3 during non-precipitation intervals), indicating increased trading activity during speed disruptions. In highly liquid markets where multiple market makers compete, the removal of HFT speed advantages primarily affects trading intensity rather than transaction costs in the post-Quincy era. When speed hierarchies erode, slower traders face lower risk of trading against better-informed counterparties, encouraging greater market participation.

For less liquid ETFs, we observe a different pattern. ES and RS decline significantly by \$0.0002 (2% of the tick size, and 1.5% of mean values in table 3), reflecting reduced trading costs for liquidity demanders and lower revenues for liquidity providers when HFT speed advantages are neutralized. These findings align with the effects observed in the pre-Quincy period and with findings in Shkilko and Sokolov (2020) and Budish et al. (2015), who suggest that leveling the speed frontier increases competition, reducing trading costs. TS increases significantly by 1.736 units (approximately 9.1% of trades reported in table 3), while VOL shows no significant change. PI declines significantly by approximately \$0.0001 (7.7% of mean values) resulting in a RV decline by \$0.00006 (9.03% of mean value). The decline in PI contrasts with expectations from studies such as Menkveld (2013), Hasbrouck and Saar (2013), and Budish et al. (2015), which highlight that reduced HFT speed diminishes their ability to mitigate adverse selection risks. Our results suggest that in less liquid markets, HFT speed advantages are associated with greater adverse selection faced by slower liquidity providers, increasing trading costs in normal (non-disrupted) periods, which is consistent with (Hendershott and Menkveld 2014). The decline in RV after controlling for volume represents an important finding: when HFT speed advantages are neutralized, prices adjust more slowly, reducing intraday volatility despite an increased number of trades. For UCO-OIL, the combination of lower spreads, lower price impacts, and lower volatility

during disruptions suggests improved market quality when speed hierarchies erode. The heterogeneity in results across ETF groups indicates that HFT impacts differ systematically across trading environments, with liquidity playing a central role. Results are robust to alternative horizons for the forward midquote ($\delta = 10, 15, 60$) as reported in table E.2 panel B.

The launch of Quincy Data reshapes HFT dynamics by expanding access to the fastest communication speeds. Before Quincy, HFTs have limited impact on highly liquid markets but increase trading costs in less liquid ones. After Quincy, slower traders in liquid ETFs may reduce participation to avoid adverse selection, lowering trading activity. In less liquid markets, HFTs cause reduced activity and increased adverse selection and volatility, prompting market makers to widen ES. This heterogeneity aligns with the notion that HFTs target ETF products based on liquidity characteristics and employ different trading strategies across products, as emphasized by Brogaard et al. (2015) and Baron et al. (2018). Our results also suggest that despite the availability of Quincy Data, speed differences persist. Our results contrast with Shkilko and Sokolov’s (2020) suggestion that Quincy erased speed differences for a sample of highly traded ETFs, mostly tracking stocks, treasury interest rates, and metals.

Precipitation disruptions to microwave networks during this period drive increases in trading activity across both ETF groups. These findings raise the question: How does the information content of trades change when HFT speed advantages are neutralized? To address this, we analyze the information shares (IS) of trades using the Hasbrouck (1991) methodology, as described in subsection 2.4 and appendix B.

Table 4. Regression Results for BP(ITU)*

Panel A: 2011-2012 USO-UNG	ES	RS 5s	PI 5s	RV	Ln(Bid MD)	Ln(Ask MD)	Ln(VOL)	NBBO	TS
BP(ITU)*	-4.65e-5 (3.71e-5)	-0.0001** (6.25e-5)	8.64e-5 (4.61e-5)	2.87e-5 (0.0001)	0.0664 (0.0001)	0.0664 (0.0392)	0.0345 (0.0467)	30.36 (20.04)	3.203 (1.953)
Lag Ln(VOL)	-2.26e-5 (1.91e-5)	-4.82e-5 (3.27e-5)	5.02e-5*** (1.83e-5)	6.41e-5*** (2.39e-5)	0.1750*** (0.0040)	0.1750*** (0.0040)		128.9*** (2.712)	17.13*** (0.4882)
Lag RV	0.0039** (0.0014)	0.0043** (0.0018)	-0.0009 (0.0011)		-3.133*** (0.4343)	-3.124*** (0.4341)	1.660*** (0.5211)	-257.2 (207.1)	163.3*** (45.54)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)
Observations	148,374	148,374	148,374	148,374	148,374	148,374	148,383	148,374	148,374
R2	0.002	0.001	0.004	0.001	0.058	0.058	0.022	0.300	0.089
Adj. R2	0.002	0.001	0.004	0.001	0.058	0.058	0.022	0.300	0.089
F-statistic	22.9616	10.0561	44.0618	9.1553	660.1867	659.9952	280.8963	4595.2715	1054.3247
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Panel A: 2011-2012 UCO-OIL	ES	RS 5s	PI 5s	RV	Ln(Bid MD)	Ln(Ask MD)	Ln(VOL)	NBBO	TS
BP(ITU)*	-0.0021*** (0.0007)	-0.0002 (0.0005)	0.0008 (0.0022)	4.56e-5 (0.0008)	-0.0458 (0.0440)	-0.0451 (0.0440)	0.0728 (0.0507)	-9.585 (21.16)	0.5332 (0.6065)
Lag Ln(VOL)	-0.0023*** (0.0005)	-0.0009*** (0.0002)	-0.0010* (0.0004)	-1.24e-5 (5.59e-5)	0.1712*** (0.0055)	0.1712*** (0.0055)		54.27*** (2.279)	3.796*** (0.1122)
Lag RV	0.4571** (0.1851)	0.0493*** (0.0177)	0.4314** (0.1855)		-3.123*** (0.3279)	-3.144*** (0.3298)	-0.5448 (0.3422)	1,023.2*** (234.7)	-3.371 (4.139)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)
Observations	119,184	119,184	119,184	119,184	119,184	119,184	119,160	119,184	119,184
R2	0.006	0.007	0.004	0.014	0.137	0.137	0.116	0.148	0.129
Adj. R2	0.006	0.007	0.004	0.014	0.137	0.137	0.116	0.147	0.129
F-statistic	66.4313	73.8600	44.3112	167.2482	1618.2669	1624.9585	1558.0933	1768.6256	1518.7144
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Panel B: 2013-2017 USO-UNG	ES	RS 5s	PI 5s	RV	Ln(Bid MD)	Ln(Ask MD)	Ln(VOL)	NBBO	TS
BP(ITU)*	-3.29e-5 (3.66e-5)	-6.38e-5 (3.94e-5)	1.08e-5 (1.59e-5)	-9.24e-8 (1.07e-5)	0.0197 (0.0282)	0.0197 (0.0282)	0.1284*** (0.0319)	4.019 (6.148)	2.893*** (1.038)
Lag Ln(VOL)	-4.91e-5*** (1.01e-5)	-7.04e-5*** (1.03e-5)	-1.14e-5*** (1.96e-6)	7.82e-6*** (2.19e-6)	0.1847*** (0.0021)	0.1846*** (0.0021)		61.07*** (0.8658)	13.48*** (0.1998)
Lag RV	0.6296*** (0.1910)	0.6176*** (0.1893)	0.0230** (0.0110)		-50.33*** (18.18)	-50.31*** (18.17)	12.76 (7.195)	5,099.4** (2,437.3)	1,008.0** (473.4)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)
Observations	513,324	513,324	513,324	513,324	513,324	513,323	513,366	513,324	513,324
R2	0.050	0.039	0.001	0.003	0.232	0.232	0.062	0.192	0.140
Adj. R2	0.050	0.039	0.001	0.003	0.232	0.232	0.062	0.192	0.140
F-statistic	2033.4640	1545.3077	36.8270	130.1423	11561.4324	11561.5380	2959.2219	9067.2451	6206.7694
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Panel B: 2013-2017 UCO-OIL	ES	RS 5s	PI 5s	RV	Ln(Bid MD)	Ln(Ask MD)	Ln(VOL)	NBBO	TS
BP(ITU)*	-0.0002** (0.0001)	-0.0002** (0.0001)	-9.23e-5** (4.56e-5)	-6.32e-5*** (2.39e-5)	-0.0257 (0.0378)	-0.0262 (0.0381)	0.0676 (0.0386)	2.162 (7.499)	1.736*** (0.6274)
Lag Ln(VOL)	-0.0007*** (2.36e-5)	-0.0006*** (1.88e-5)	-0.0002*** (1.01e-5)	8.99e-7 (4.54e-6)	0.2419*** (0.0031)	0.2410*** (0.0031)		56.51*** (0.9486)	6.656*** (0.0918)
Lag RV	1.000*** (0.3716)	0.5566** (0.1830)	0.0594** (0.0198)		-33.10*** (5.752)	-33.51*** (5.837)	0.1190 (1.266)	234.0 (339.6)	87.75* (41.60)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)
Observations	436,204	436,204	436,204	436,204	436,204	436,192	436,296	436,204	436,204
R2	0.039	0.030	0.001	0.001	0.096	0.096	0.093	0.227	0.139
Adj. R2	0.039	0.030	0.001	0.001	0.096	0.096	0.093	0.227	0.139
F-statistic	1455.6432	1120.1624	48.6257	33.6370	3845.8317	3824.3937	4311.1455	10573.7497	5823.6027
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Note: This table presents panel regression results from equation (1), estimating the impact of BP(ITU)* precipitation disruptions on market quality measures. The analysis uses minute-level data, incorporates fixed effects to control for security-specific and intraday patterns, and applies Driscoll-Kraay standard errors to account for cross-sectional dependence and autocorrelation (with L in parentheses indicating the number of lags used to model error term correlation). Lagged controls for volume and realized volatility are included to address potential endogeneity. Definitions and summaries of the market quality measures are provided in Section 2.4 and Table 2.

3.2 Price Discovery Results

We compute the IS of trades following Hasbrouck (1991), for three distinct periods: immediately before, during, and immediately after precipitation events, using millisecond DTAQ data from May 2013 to 2017 for each market. As discussed in Section 2.4., our approach accounts for the continuous process of price discovery, making precise event timing important. We compare the trade-related IS during disruptions to non-disruption (combined before–after) periods and test the following hypotheses:

$$H0: \text{median}(IS_{\text{trade,combined}}) - \text{median}(IS_{\text{trade,during}}) = 0 \quad (2)$$

$$H1: \text{median}(IS_{\text{trade,combined}}) - \text{median}(IS_{\text{trade,during}}) < 0 \quad (3)$$

where $IS_{\text{trade,combined}}$ represents the average information share of trades before and after disruptions, and $IS_{\text{trade,during}}$ represents the information share during disruptions. Our alternative hypothesis suggests that trades become more informative when speed advantages are temporarily neutralized. This may occur because the fastest trades introduce more noise into the price discovery process, reducing the information content of individual transactions (Fleming et al. 2018; Ma and Serra 2024). In a more even playing field, trades may carry greater information content, which is consistent with the increased trading activity during disruptions documented in Section 3.1.

We illustrate our results in Figure 3, which compares the distributions of trade information shares for the combined and disruption periods. These probability densities are generated using the Epanechnikov kernel, which is well-suited for capturing distributional differences. The Wilcoxon rank-sum (unpaired) test is significant at the 5% level across all ETFs, while the signed-rank test (paired) is statistically significant for USO and marginally significant ($p < 0.1$) for UNG. The unpaired test evaluates whether the overall distribution of IS during disruptions differs from that of the combined before–after periods, whereas the

paired test compares changes within matched intervals, assessing whether IS systematically rises or falls during disruptions relative to its own baseline.

Results reveal substantial heterogeneity across liquidity groups. For the more liquid USO–UNG pair, trades are, on average, 13.41% more informative during disruptions. For the less liquid UCO–OIL pair, trades are only 0.63% more informative, indicating only a marginal shift. The results suggest that HFT speed advantages disproportionately reduce trade informativeness in highly liquid markets, where the fastest traders’ high-frequency transactions appear to introduce noise rather than fundamental information into the price discovery process.

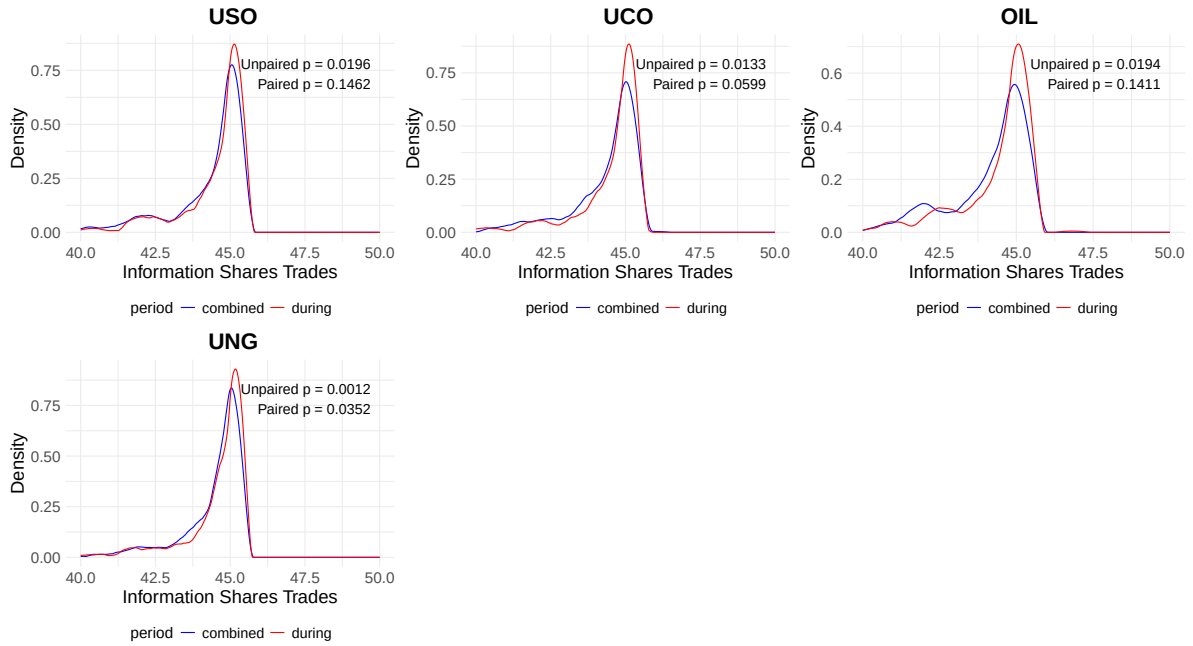


Fig. 3. 2013-2017 Price Discovery Share Distributions

Note: This figure illustrates the distributions of the Hasbrouck (1991) price discovery information shares (IS) of trades with probability densities derived using the Epanechnikov kernel for the trade components of our sample of energy ETFs for the May 2013-2017 period post Quincy Data launch. We use $P(ITU)^*$ to determine disruption intervals. The mathematical details are described in appendix B. We present the Wilcoxon test p-values (Wp) of the following hypothesis tests in the upper right side of each commodity-related security:

$$H0: \text{median}(IS_{\text{trade,combined}}) - \text{median}(IS_{\text{trade,during}}) = 0$$

$$H1: \text{median}(IS_{\text{trade,combined}}) - \text{median}(IS_{\text{trade,during}}) < 0$$

Both unpaired and paired tests are shown for each ETF.

4 Concluding and Main Findings

High-frequency traders' pursuit of speed advantages has reshaped financial market structure, yet whether this transformation improves or degrades market quality remains contested. This study provides evidence that the market-quality effects of HFT speed depend on the trading environment, using energy ETFs as a setting where cross-market arbitrage plays a central role.

Futures trade in Chicago while ETFs trade in New York, creating a geographically fragmented market connected by microwave and fiber networks. We exploit weather-induced disruptions to these microwave networks to identify periods when HFT speed advantages are temporarily neutralized. Using ITU-based precipitation thresholds grounded in the physics of signal attenuation, Thiessen-polygon spatial aggregation, and panel data regression, we document limited HFT effects during 2011-2012, consistent with low microwave network adoption during the technology's nascent period. Substantial impacts emerge during 2013–2017 following Quincy Data's introduction of an ultra-low-latency market-data feed, which broadened access to faster intermarket information and is consistent with increased participation in spatial latency arbitrage.

In highly liquid ETFs (USO, UNG), HFT speed advantages deter market participation; removing these advantages increases trading volume by 12.8% without affecting quoted spreads. This suggests that speed hierarchies primarily discourage participation rather than raising transaction costs. In less liquid ETFs (UCO, OIL), HFT speed advantages widen quoted spreads by 1.5%, increase price impacts by 7.7%, and raise volatility by 9.03%. Removing these advantages reverses these effects and increases trade counts by 9.1%, indicating that speed advantages are associated with wider spreads and larger price impacts in markets with limited liquidity. The differential effects across liquidity regimes suggest that latency

advantages are more consequential in thinner markets. For highly liquid ETFs, trades become 13.41% more informative when speed advantages are neutralized, indicating that the fastest traders’ high-frequency transactions exploit transient price discrepancies rather than incorporate fundamental information into prices.

For policymakers and regulators, our evidence suggests that the effects of speed-based competition vary systematically with market liquidity, highlighting the potential value of market-contingent policies—such as applying speed bumps selectively to less liquid securities or implementing liquidity-threshold-based rules—rather than blanket interventions. The persistence of speed advantages even after Quincey Data expanded access to microwave technology indicates that technological solutions alone may not level the playing field. Differences in trading speed remain an important feature of market structure.

For traders and asset managers, our findings indicate that non-HFT market participants face higher adverse selection costs in less liquid securities. The increase in trading activity when speed advantages are neutralized is consistent with reduced concerns about adverse selection. While we do not observe trading strategies directly, these results suggest that trading frictions related to speed can affect participation. Institutional investors managing large positions in less liquid energy ETFs should account for the higher transaction costs documented here when evaluating portfolio rebalancing costs and the feasibility of commodity exposure strategies.

Our analysis has several important limitations. First, our sample focuses on four energy ETFs during 2011-2017; effects may differ in other commodity ETFs, time periods, or market structures. Second, our weather-based identification strategy reveals HFT impacts only during precipitation events, which may not fully represent HFT behavior under all market conditions. Third, we cannot observe individual trader identities or strategies, limiting our ability to distinguish among different types of HFT firms and trading approaches. Finally, our findings reflect the technological and regulatory environment of our sample period; ongoing changes in market structure and trading technology may alter HFT impacts over

time.

Acknowledgments

We thank [REDACTED] seminar participants for helpful discussions. We acknowledge support from the [REDACTED] Research at the [REDACTED]. We thank [REDACTED] of the CME for processing our CME Datamine orders and offering help when needed. All errors are our own.

Author Contributions

[REDACTED] Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Software; Validation; Visualization; Writing – original draft; Conceptualization.

[REDACTED] Methodology; Supervision; Writing – review and editing; Conceptualization.

Data statement

We have signed nondisclosure agreements for the financial data used in this study. All access and use of CME and NYSE DTAQ data is subject to CME and NYSE Data Terms of Use. Precipitation data from NOAA is freely available online.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used chatGPT in order to improve in some instances the sentence structure and readability of the research article and replication codes. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Appendix A

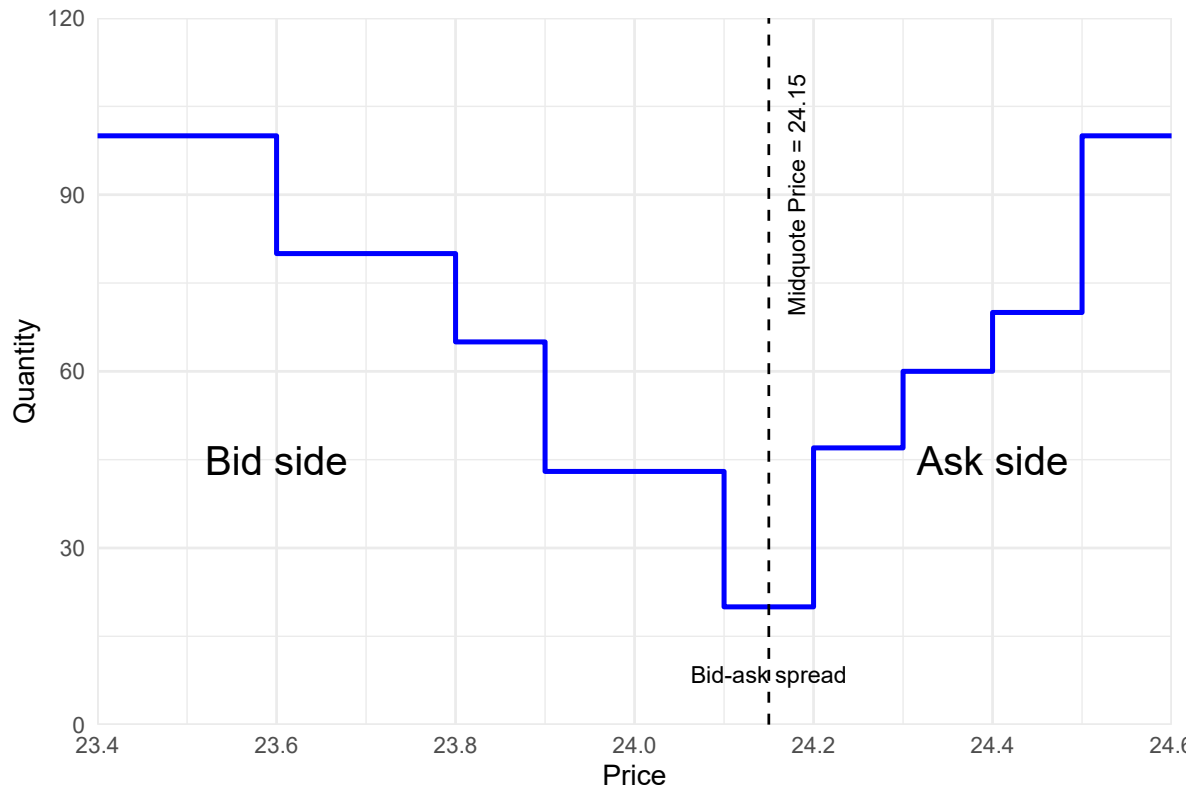


Fig. A.1. An example of a limit order book

Note: This figure displays a hypothetical limit order book (LOB) comprising the market supply, the ask side, and demand, the bid side. The horizontal axis measures the price. The vertical axis measures the cumulative quantity in number of shares. Each level in the step-wise graph represents the quantities demanded and supplied at each specific price. The bid-ask spread is 0.2 cents, which is the difference between the best ask and best bid prices. The midquote price is defined as the arithmetic mean between the best bid and ask prices, and indicated by the vertical dashed line. The tick size is defined as the minimum price increment allowed in the market, which is 0.1 cents in our example. The bid-ask spread represents the minimum transaction cost that traders need to pay to purchase a single share.

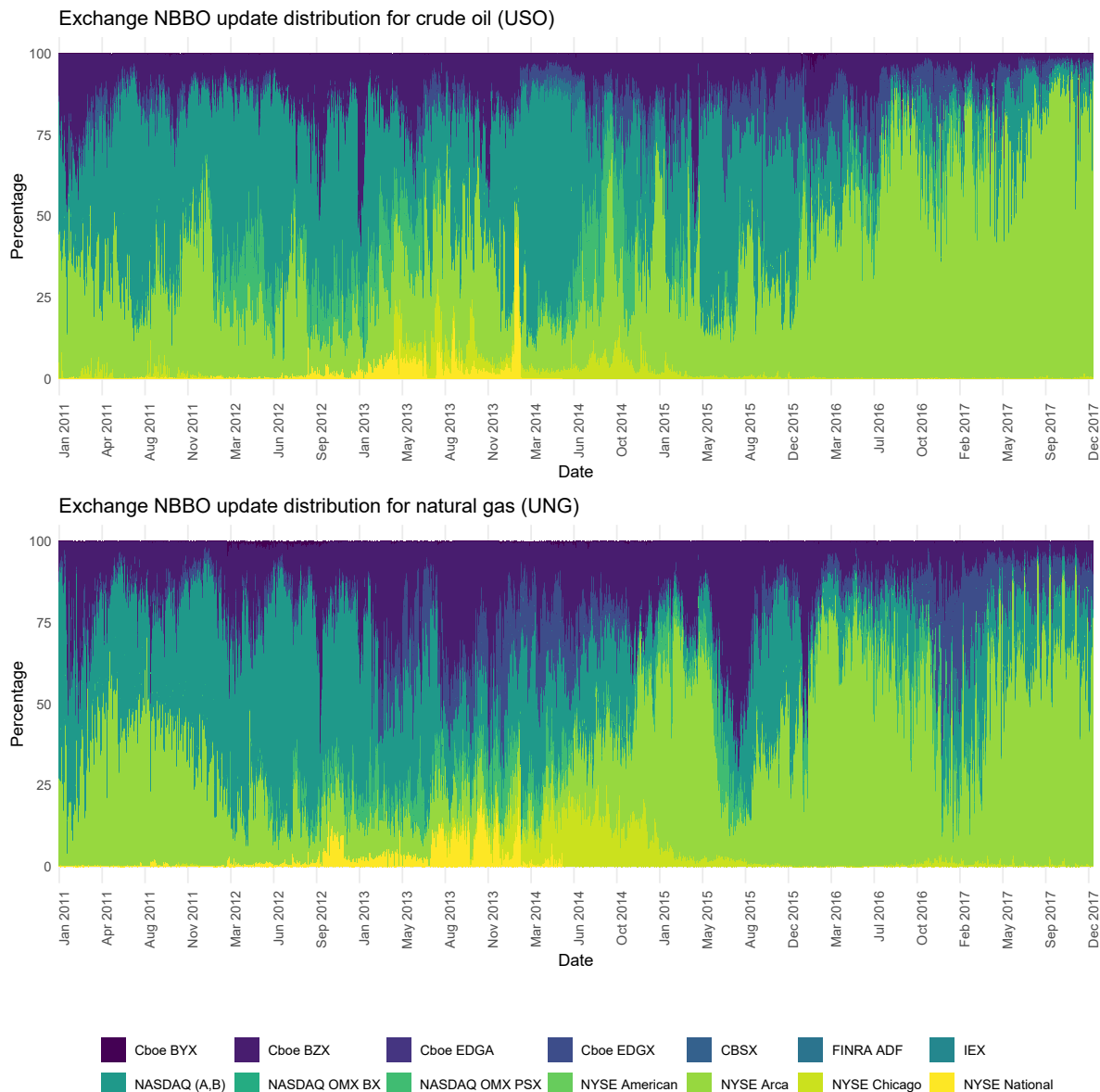


Fig. A.2. Distribution of national best bid and offer updates by exchange

Note: The figure presents the daily percentage of National Best Bid and Offer (NBBO) updates by exchange. The plots illustrate daily percentages throughout the 2011-2017 sample period for the following ETFs: United States Oil Fund (USO) and United States Natural Gas Fund (UNG). The exchanges operated by the New York Stock Exchange (NYSE) utilize the electronic matching engine located in Mahwah, New Jersey. The exchanges operated by the National Association of Securities Dealers Automated Quotations (NASDAQ) and the Chicago Board Options Exchange (Cboe) have their matching engines located in Carteret, New Jersey and Secaucus, New Jersey, respectively. Minor exchanges like the Investors Exchange (IEX) and the Financial Industry Regulatory Authority Alternative Display Facility (FINRA ADF) have engines located in Weehawken, New Jersey and Trumbull, Connecticut, respectively (FINRA ADF closed on October 2019).

Table A.1. An example of daily trade and quote (DTAQ) data, quotes

Date	Time	Bid	Ask	Bid Size	Ask Size	Quote Seq. #	NBBO	Ticker	Exchange
2013-10-16	8:18:43.381	19.65	19.78	50	1	32040	6	UNG	P
2013-10-16	8:18:43.381	19.65	19.78	50	51	32041	0	UNG	P
2013-10-16	8:18:45.323	19.65	19.78	50	1	32073	0	UNG	P
2013-10-16	8:18:45.326	19.66	19.78	50	1	32074	6	UNG	P
2013-10-16	8:18:45.327	19.65	19.78	50	1	32075	6	UNG	P
2013-10-16	8:18:45.328	19.66	19.78	50	1	32076	6	UNG	P
2013-10-16	8:18:45.329	19.65	19.78	50	1	32077	6	UNG	P
2013-10-16	8:18:48.088	19.65	19.78	50	51	32092	0	UNG	P
2013-10-16	8:18:48.098	19.64	19.78	50	51	32093	6	UNG	P
2013-10-16	8:18:48.191	19.65	19.78	50	51	32094	6	UNG	P

Note: This table provides an example of the Daily Trade and Quote (DTAQ) dataset for United States Natural Gas Fund (UNG). DTAQ provides trade and quote information issued from NYSE markets and the U.S. Consolidated Tape covering all U.S. Equities instruments. The Date and Time columns specify the timestamp of each quote update to the millisecond. The Bid and Ask columns record the respective bid and ask prices at the moment of the update, while the Bid Size and Ask Size indicate the corresponding quantities available at those prices. The Quote Sequence # uniquely identifies each quote update within the dataset. The NBBO column denotes whether the quote update contributes to the National Best Bid and Offer (6 indicates a new update to the NBBO, while 0 indicates no new update). The Ticker specifies the security as UNG, and the Exchange column identifies the exchange where the quote update originated.

Table A.2. An example of daily trade and quote (DTAQ) data, trades

Date	Time	Price	Size	Trade Seq. #	Ticker	Exchange
2013-10-16	7:58:31.655	19.74	200	444	UNG	P
2013-10-16	8:20:01.029	19.65	600	655	UNG	K
2013-10-16	8:23:37.083	19.74	500	703	UNG	K
2013-10-16	8:24:11.374	19.75	500	707	UNG	K
2013-10-16	8:35:36.294	19.74	500	865	UNG	K
2013-10-16	8:36:25.137	19.73	500	875	UNG	J
2013-10-16	8:47:52.795	19.70	1000	987	UNG	K
2013-10-16	8:47:52.807	19.70	2000	988	UNG	K
2013-10-16	9:03:06.298	19.76	100	1437	UNG	P
2013-10-16	9:03:13.301	19.75	150	1441	UNG	D

Note: This table presents an excerpt from the Daily Trade and Quote (DTAQ) dataset for United States Natural Gas Fund (UNG), showcasing trade updates across U.S. equities. The Date and Time columns provide the timestamp for each trade, recorded to the millisecond. The Price column specifies the transaction price of the trade, while the Size column captures the number of shares traded. The Trade Sequence # uniquely identifies each trade update within the dataset. The Ticker column identifies the traded security as UNG, and the Exchange column denotes the exchange where the trade occurred. While the table does not include a direct indicator for the National Best Bid and Offer (NBBO), most trades typically align with the NBBO due to regulatory guidelines like SEC Rule 611 (Order Protection Rule). However, trades can sometimes occur outside the NBBO due to factors such as price improvement, hidden orders, or market conditions.

Table A.3. An example of CME time and sales data

Date	Time	Price	Size	Trade Seq. #	Ticker	Exchange
2013-10-16	7:58:31.655	19.74	200	444	UNG	P
2013-10-16	8:20:01.029	19.65	600	655	UNG	K
2013-10-16	8:23:37.083	19.74	500	703	UNG	K
2013-10-16	8:24:11.374	19.75	500	707	UNG	K
2013-10-16	8:35:36.294	19.74	500	865	UNG	K
2013-10-16	8:36:25.137	19.73	500	875	UNG	J
2013-10-16	8:47:52.795	19.70	1000	987	UNG	K
2013-10-16	8:47:52.807	19.70	2000	988	UNG	K
2013-10-16	9:03:06.298	19.76	100	1437	UNG	P
2013-10-16	9:03:13.301	19.75	150	1441	UNG	D

Note: This table presents an example of CME Time and Sales data for Natural Gas Futures. The Date column specifies the calendar day of the trade, while the Time column provides the second the trade occurred, with ties indicating multiple trades executed at the same second. The Price column shows the execution price, and the Size column details the number of contracts involved in the trade. The Trade Seq. # column assigns a unique identifier to each trade for sequential tracking. Finally, the Contract column identifies the specific futures contract being traded, such as NGF16 for January 2016 Natural Gas Futures.

Appendix B

This appendix provides details on the price discovery analysis performed using the Chicago futures and NY energy ETFs, aimed at determining the direction of price signals that are arbitrated by HFTs using microwave transmissions. We implement both the Hasbrouck (1995) information shares (IS) and the Putniņš (2013) information leadership shares (ILS) methods. By relying on the US NBBO and consolidated trade file data, we treat the NY cluster as a single market and compare it with the futures market. The following sections provide the mathematical details and results for the price discovery model.

Market events such as trades or limit orders, provide valuable information that HFTs may use for spatial latency arbitrage. Following previous research (Shkilko and Sokolov 2020), we hypothesize that spatial arbitrage follows the direction of market information flows, which we investigate. We evaluate the direction of information flows between CME futures and related equities traded in the NY cluster using both trading volume and Putniņš (2013) ILSs to identify the markets where HFTs are likely to capitalize on information. For instance, if information predominantly flows from futures to related ETF markets, spatial arbitrage algorithms are likely to have a larger effect on equity market quality.

Trading volumes can provide insights into how information is impounded into the market, with higher volumes potentially indicating more informative markets compared to lower volumes (Hasbrouck 1995).¹ Figure B.1 presents the logarithmic daily traded dollar volume for our sample of Chicago futures and energy ETFs traded in the NY cluster from 2011-2017, using the front-month rolling method for futures.² The graph illustrates that futures volume typically exceeds that of the NY cluster ETFs, except in the days preceding the futures rollover from the old contract to the next-to-expire contract, when the volume may dip below that of the corresponding NY cluster security. The larger futures trading volume

¹Limit orders can significantly contribute to price discovery (Fleming, Mizrach, and Nguyen 2018; Ma and Serra 2024). Because our objective is to compare information impounded across different markets, rather than through different market events, we simplify and focus on trades.

²Bloomberg builds its front-month series by rolling contracts to the next-to-expire around the fifth business day of the month prior to expiration.

suggests that information flows from Chicago to the NY cluster.

We calculate Putniņš (2013) information leadership shares (ILS) using five-second calendarization, which involves taking the last transaction price within each five-second interval during trading hours. ILS are robust to differing noise levels across markets. This appendix also presents robustness checks regarding price discovery measures and calendarization. Price discovery shares are derived from pairwise models that match each CME commodity futures with a corresponding NY ETF. We calculate price discovery shares for energy commodities (crude oil and natural gas).

B.1 Price discovery (1995) mathematical details

Let $p_{1,t}, p_{2,t}$ represent the natural logarithm transformation of transaction prices of *Futures* and *ETF* respectively. Let the cointegration relationship be $p_{1,t-1} - p_{2,t-1} + \mu = ECT_{t-1}$ with μ denoting a restricted constant and ECT_{t-1} denoting the error correction term, measuring deviations from the equilibrium parity. The vector error correction model (VECM) is then specified as:

$$\begin{aligned}\Delta p_{1,t} &= \alpha_1(p_{1,t-1} - p_{2,t-1} + \mu) + \sum_{i=1}^l \gamma_i \Delta p_{1,t-i} + \sum_{i=1}^l \delta_i \Delta p_{2,t-i} + \varepsilon_{1,t} \\ \Delta p_{2,t} &= \alpha_2(p_{1,t-1} - p_{2,t-1} + \mu) + \sum_{i=1}^l \psi_i \Delta p_{1,t-i} + \sum_{i=1}^l \phi_i \Delta p_{2,t-i} + \varepsilon_{2,t}\end{aligned}\tag{1}$$

The VECM covariance matrix is given by

$$\Omega = (\omega_1, \omega_2)\tag{2}$$

Let $\omega_1 = (\sigma_1^2, \rho\sigma_1\sigma_2)^\top$ and $\omega_2 = (\rho\sigma_1\sigma_2, \sigma_2^2)^\top$. Consider the Cholesky factorization $\Omega =$

$MM^\top$, where $M = (m_1, m_2)$ with

$$m_1 = (m_{11}, m_{12})^\top = (\sigma_1, \rho\sigma_2)^\top, \quad m_2 = (0, m_{22})^\top = (0, \sigma_2\sqrt{1-\rho^2})^\top.$$

The Information Share (IS) and the Component Share (CS) are defined as

$$CS_1 = \nu_1 = \frac{\alpha_2}{\alpha_2 - \alpha_1}, CS_2 = \nu_2 = \frac{\alpha_1}{\alpha_1 - \alpha_2} \quad (3)$$

$$IS_1 = \frac{(\nu_1 m_{11} + \nu_2 m_{12})^2}{(\nu_1 m_{11} + \nu_2 m_{12})^2 + (\nu_2 m_{22})^2}, IS_2 = \frac{(\nu_2 m_{22})^2}{(\nu_1 m_{11} + \nu_2 m_{12})^2 + (\nu_2 m_{22})^2} \quad (4)$$

Because the estimates of IS are affected by the ordering of the price series in the VECM, following Putniņš (2013) we calculate IS under each of the two possible orderings and then take the simple average.

The Yan-Zivot information leadership metric is given by:

$$IL_1 = \left| \frac{IS_1 CS_2}{IS_2 CS_1} \right|, IL_2 = \left| \frac{IS_2 CS_1}{IS_1 CS_2} \right| \quad (5)$$

The information leadership shares (ILS) are given by:

$$ILS_1 = \frac{IL_1}{IL_1 + IL_2}, ILS_2 = \frac{IL_2}{IL_1 + IL_2} \quad (6)$$

B.2 Price discovery (1995) results

We perform a price discovery analysis using both Hasbrouck (1995) IS and Putniņš (2013) ILS methods using a 5 second calendarization, the maximal frequency recommended in O'Hara (2015). Additionally, we consider 1 second calendarization, the maximum frequency available from the CME Time and Sales data. We use the Johansen (1991) cointegration test to determine when the Chicago futures and NY ETF are cointegrated. We exclude days where prices are either stationary or nonstationary without cointegration from the price

discovery analysis.

The graphs present the information shares for the CME futures. The price discovery shares of NY ETF can be calculated as one minus the computed Chicago futures price discovery share. Figure B.2 Panel A shows ILS shares for CL with respect to OIL using a 5 second calendarization, panel B shows ILS shares for CL with respect to UCO using a 5 second calendarization, panel C shows ILS shares for CL with respect to USO using a 1 second calendarization, and panel D shows IS shares for NG with respect to UNG using a 1 second calendarization. Using different price discovery methods with varying calendarizations produces results consistent with futures markets dominating price discovery over their security counterparts. This ensures that results reflect true market dynamics rather than methodological or temporal artifacts.

Tables B.1 and B.2 present additional details on the daily cointegration analysis for the CL-USO and NG-UNG price series pairs using both 1 and 5 second calendarization. The tables presents summary statistics for Johansen cointegration test results and the speed of adjustment coefficients α 's in equation (1) of this appendix. We find results to be qualitatively similar across calendarizations. Most days show cointegration (about 1500 out of 1761). In the main text, we rely on the 5 second calendarization ILS to determine the information flow of price signals. However, results are robust to different calendarizations and methodologies.

B.3 Price discovery (1991), the informativeness of trades mathematical details

To determine whether spatial latency arbitrage activities alter the information content of ETF trades, we employ a within-security price discovery approach based on Hasbrouck (1991). This allows us to estimate how trades contribute to incorporating information into the efficient market price. Previous literature relies on the midpoint price as the efficient market price observed with noise (e.g., Hasbrouck 1991; Lin, Lo, and Qiao 2021). We

estimate a Structural Vector Autoregressive (SVAR) model of order p using ordinary least squares (OLS):

$$\mathbf{A}_0 \mathbf{y}_t = \sum_{i=1}^p \mathbf{A}^* \mathbf{y}_{t-i} + \mathbf{e}_t \quad (7)$$

where $\mathbf{e}_t \sim \text{i.i.d.}(\mathbf{0}, I)$ is the vector of structural shocks, with $\mathbf{0}$ a 5×1 vector of zeros and I the 5×5 identity matrix. The 5×1 endogenous variable vector is

$$\mathbf{y}_t = [r_t, x_t, xv_t, (xv_t)^2, s_t]^\top.$$

It contains midpoint returns $r_t = m_t - m_{t-1}$ representing efficient-price returns. Trade characteristics are captured by trade direction x_t (+1 for buys and -1 for sells), signed trade volume xv_t measured in shares, and squared volume $(xv_t)^2$ to allow for nonlinear impacts of volume on midpoint prices. Trade costs are reflected by the bid–ask spread s_t , which also provides information about perceived information asymmetry. While not all trades are informed, the perception that a trade is informed can generate adverse selection risk and widen spreads. Variables in $\mathbf{y}_t$ are measured in event time, where events are trades. The SVAR order p is selected using the Bayesian information criterion with a maximum of 10 lags.

The matrix $\mathbf{A}_0$ is composed of structural parameters that show the contemporaneous relationship between the endogenous $\mathbf{y}_t$ variables. This matrix is key for the identification of the SVAR model, as it defines how the structural innovations drive the variables in the model. Matrix $\mathbf{A}_0$ is defined based on the mechanics of the LOB described in the next section of this appendix as follows:³

³It is important to note that our identification strategy is dictated by the mechanics of the LOB and does not depend on assumptions regarding the direction of causality. This differs from other price discovery analyses based on vector error-correction models that require assumptions on the direction of causality given the lack of economic theory (see Putniņš 2013).

$$\mathbf{A}_0 = \begin{pmatrix} 1 & -\alpha_{12} & -\alpha_{13} & -\alpha_{14} & -\alpha_{15} \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (8)$$

Parameters in the first row capture the contemporaneous impacts of trade variables and spreads on midpoint returns. Both trades and quote revisions may contemporaneously change midpoint returns. For example, a trade that buys all ask quotes at the top of the book will automatically increase the spread and the midpoint price. A cancellation of all the ask quotes available at the top of the book will yield the same effect. The first column in matrix $\mathbf{A}_0$ reflects that midpoint returns do not have a contemporaneous impact on trades and spreads.

Following Hasbrouck (1991) and more recent literature (Chaboud, Hjalmarsson, and Zikes, 2021; Ma and Serra 2024), we calculate information shares (IS) by separating the permanent component of the midpoint price from the transitory pricing errors. The permanent price component is identified through impulse response analysis (IRF), which measures the impacts of a market event on the midpoint price over time. The IRF is derived from the Vector Moving Average (VMA) representation of the SVAR model:

$$\mathbf{y}_t = \sum_{i=1}^{\infty} \Psi_i \mathbf{e}_{t-i} \quad (9)$$

where Ψ_j are MA terms. Beveridge and Nelson (1981) decompose this as follows:

$$\mathbf{y}_t = \psi(\mathbf{1})\mathbf{e}_t + \eta_t \quad (10)$$

where $\psi(\mathbf{1}) = \sum_{i=1}^{\infty} \Psi_i$ is the long-run MA matrix. In terms of the midpoint returns equation, $\psi(\mathbf{1})\mathbf{e}_t$ provides the returns of the permanent price process and η_t measures the temporary

pricing errors. Let the first-row elements of $\Psi(\mathbf{1})$ be denoted as ψ :

$$\psi = [\psi_1, \psi_2, \psi_3, \psi_4, \psi_5] \quad (11)$$

The ISs are derived based on the variance of the permanent component in $\mathbf{y}_t$. After a shock on variable j , the ISs are expressed as:

$$IS_j = \frac{\psi_j^2 \sigma_j^2}{\psi \Omega \psi^\top}. \quad (12)$$

where σ_j^2 is the variance of the j -th structural shock and Ω is the covariance matrix of the structural shocks.

By comparing ISs generated during precipitation disruption events to those generated during non-disrupted intervals, we assess how the removal of speed advantages affects the information content of trades. To capture the combined information content of different elements of trade (sign, volume, and square volume), we derive a composite share as: $IS_{\text{trade}} = IS_2 + IS_3 + IS_4$. Notice that IS_{trade} excludes the autoregressive effect of r_t , which captures the lag impacts of previous market activity that cannot be attributed to the current trades.

Price discovery is a continuous process that varies with public and private information, often differing across days as this information changes. To analyze how this continuous process shifts due to HFT speed advantages, we estimate the information shares separately for three distinct periods relative to each identified rain interval; before, during and after. The before and after periods correspond to the 15 minutes immediately preceding and following a precipitation disruption interval. The during period covers the minutes corresponding to the precipitation disruption interval. For each of these periods, we estimate a separate SVAR model using event time and compute the corresponding information shares as described above. To isolate the effect of precipitation, we also compute a combined "no disruption" measure by averaging the information shares from the "before" and "after" periods:

$$IS_{\text{trade,combined}} = \frac{IS_{\text{trade,before}} + IS_{\text{trade,after}}}{2} \quad (13)$$

where $IS_{\text{trade,before}}$ and $IS_{\text{trade,after}}$ are the information shares for variable j in the before and after periods respectively. Since IS values are non-normally distributed according to the Jarque-Bera test, we compare $IS_{\text{trade,during}}$ with $IS_{\text{trade,combined}}$ using non-parametric statistical tests, specifically the paired and unpaired Wilcoxon signed-rank tests. This comparison helps determine whether there are significant changes in the information content of trades during precipitation events, which may suggest shifts in the behavior of HFTs when speed advantages are temporarily reduced. By applying this methodology across precipitation events over our entire sample period, we can understand how precipitation-induced disruptions to HFTs activity affect the price discovery process in our sample of energy ETFs.

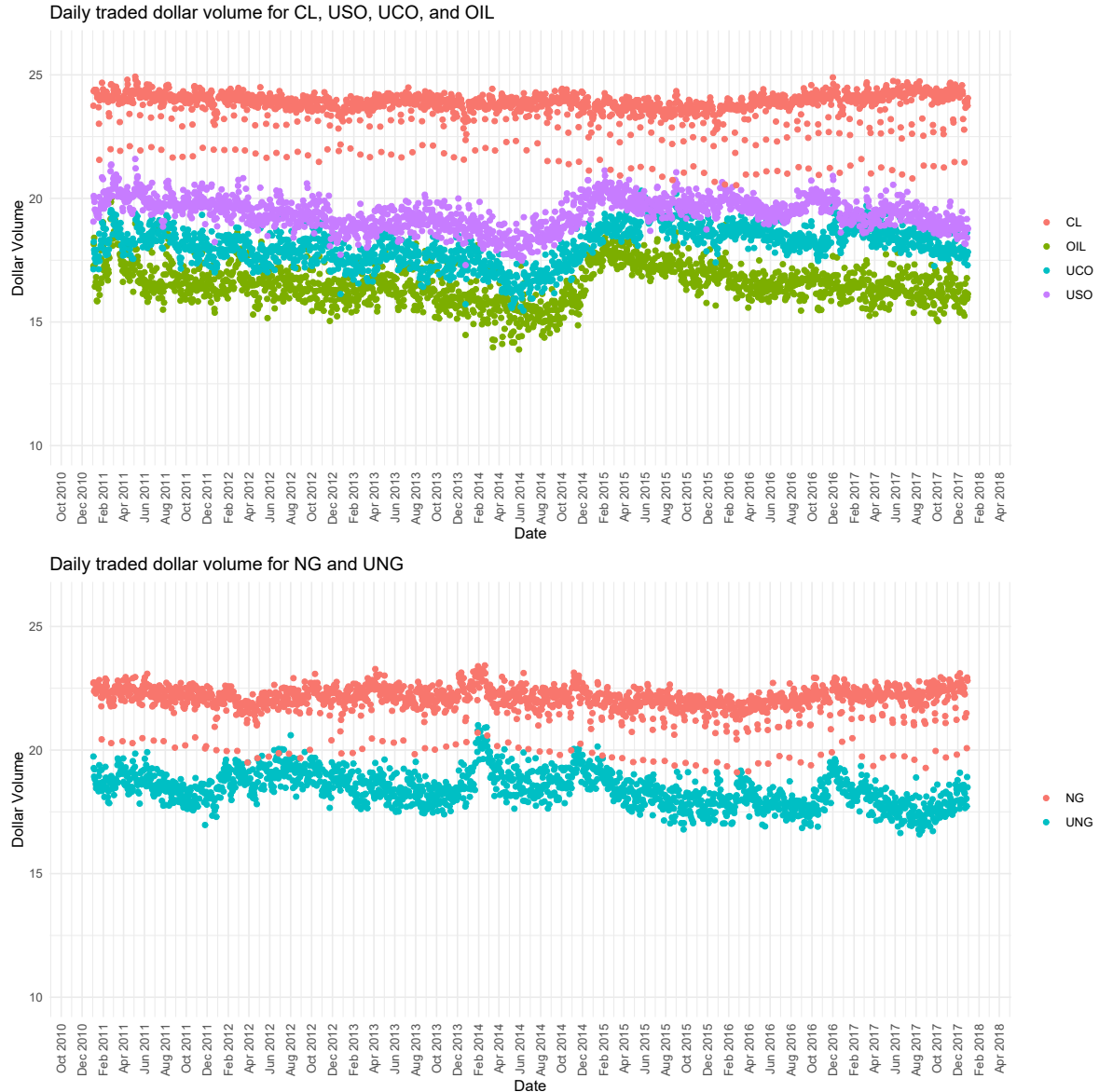


Fig. B.1. Daily traded dollar volume for related Chicago futures and New York securities
 Note: The figure presents the logarithmic daily traded dollar volumes of various financial instruments over time, divided into six distinct plots. The first plot illustrates the trading activity for Crude Oil Futures (CL), United States Oil Fund (USO), ProShares Ultra Bloomberg Crude Oil (UCO), and iPath Pure Beta Crude Oil ETN (OIL). The second plot focuses on Natural Gas Futures (NG) and the United States Natural Gas Fund (UNG). The third plot displays the S&P500 E-mini futures (ES) and SPDR S&P 500 ETF Trust (SPY). The logarithmic scale is used to manage the wide range of trading volumes effectively. Futures contracts are rolled using the front-month method.

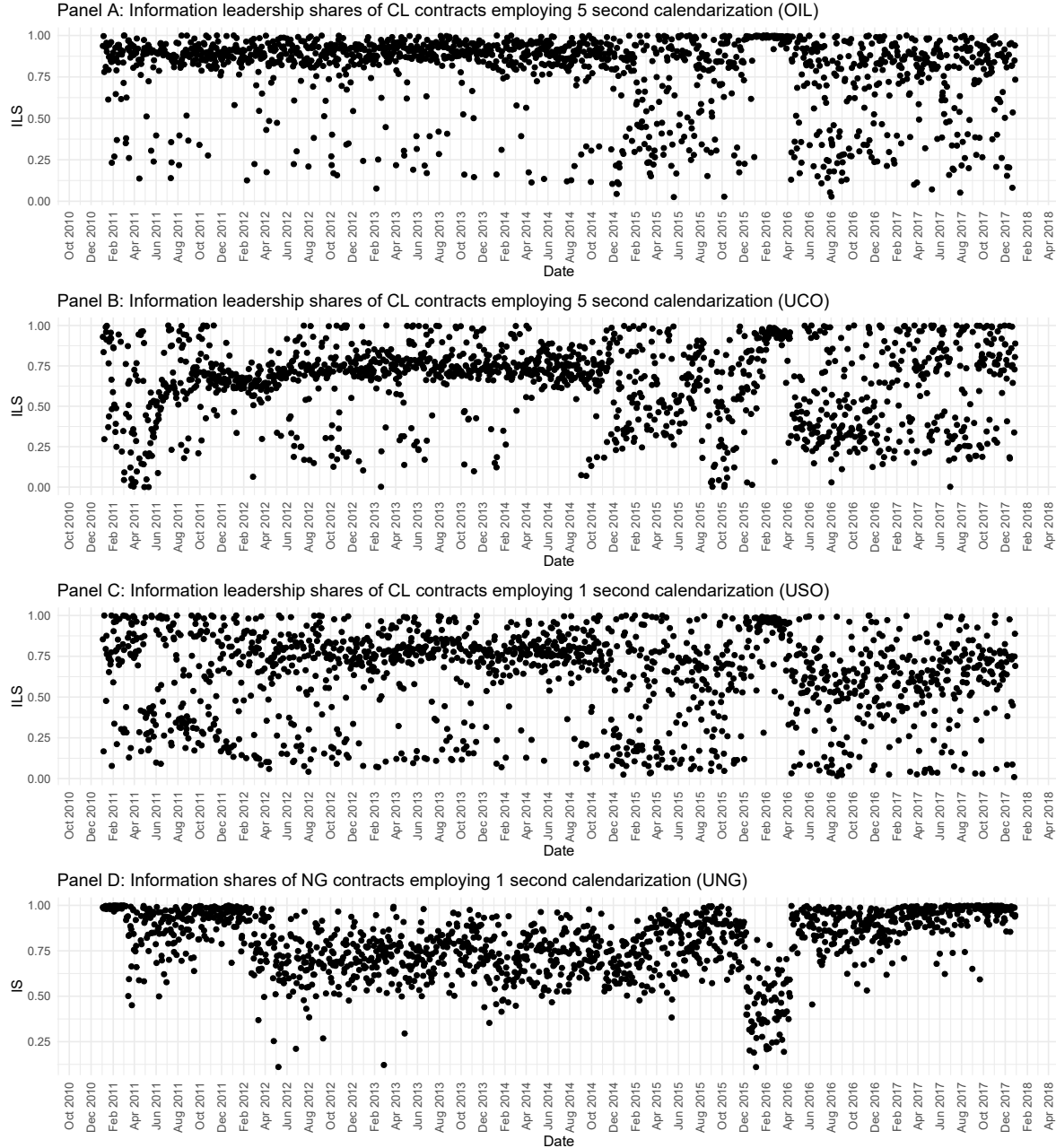


Fig. B.2. Daily price discovery share results derived using various calendarizations and methods

Note: This figure presents Putniņš (2013) Information Leadership Shares (ILS) and Hasbrouck (1995) Information Shares (IS) for West Texas Intermediate Crude Oil (CL) and Henry Hub Natural Gas (NG) Futures contracts traded at the Chicago Mercantile Exchange. CL is paired with ProShares Ultra Bloomberg Crude Oil (UCO), United States Oil Fund (USO) and the iPath Pure Beta Crude Oil ETN (OIL). NG is paired with the United States Natural Gas Fund LP (UNG). Since these shares sum to one for every pair, we present only futures price discovery shares. The calendarization frequency and price discovery method are specified in the title of each panel. The price discovery measure is indicated in the vertical axis.

Table B.1. Summary statistics by calendarization frequency for CL-USO

	1 sec	5 sec
Total Days	1761	1761
No Cointegration or Stationary Days	250	242
No Cointegration Days	246	203
Stationary Days	4	39
Cointegration Days	1511	1519
α USO p value > 0.05	1492	1479
α CL p value > 0.05	727	845
Both α 's p values > 0.05	723	841

Note: This table presents summary statistics for the Johansen cointegration test for the NG-UNG price series. We also present the p-values for the speed of correction coefficients, namely, the α 's in equation (1) of this appendix. We show results for two different sampling frequencies, 1 and 5 seconds.

Table B.2. Summary statistics by calendarization frequency for NG-UNG

	1 sec	5 sec
Total Days	1761	1761
No Cointegration or Stationary Days	260	248
No Cointegration Days	247	196
Stationary Days	13	52
Cointegration Days	1501	1513
α UNG p value > 0.05	483	703
α NG p value > 0.05	1296	1111
Both α 's p values > 0.05	462	645

Note: This table presents summary statistics for the Johansen cointegration test for the NG-UNG price series. We also present the p-values for the speed of correction coefficients, namely, the α 's in equation (1) of this appendix. We show results for two different sampling frequencies, 1 and 5 seconds.

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Appendix C

This appendix provides details on the precipitation data and procedures used in this research. The next two sections discuss the mathematical details for aggregating regional precipitation using the arithmetic mean and Thiessen polygon methods. Then, we describe the procedure that calculates a spherical quadrangle given two points. Following this, we analyze and discuss the precipitation data sets.

We use precipitation to identify HFT activities as precipitation attenuates the strength of the radio signals passing through rain or snow. When a signal travels through precipitation, its energy is partially absorbed and scattered by raindrops or snowflakes. This affects the quality and reliability of wireless communications, especially at higher frequencies.

NOAA provides accumulated precipitation data, measured in inches every 15-minutes, through the publicly available dataset DSI-3260, covering the years 2011 to 2013. For 2014 and onward, NOAA provides the alpha version of the 15-minute observations used to compute the Hourly Precipitation Dataset. The counter resets to 0 after each 15-minute reporting period. Both datasets collect precipitation using Fischer-Porter gauges housed in U.S. weather stations. The exact location of each weather station is also available from NOAA. In contrast, the exact locations of microwave networks in the Chicago – New York corridor change over time and are not readily available, challenging the selection of relevant weather stations for identifying precipitation intervals that may disrupt information transfer via microwave technology. Also, weather stations provide an imprecise indicator of disruptions to HFT firm activities, since precipitation at a particular station does not necessarily affect the microwave paths used by HFTs.

Shkilko and Sokolov (2020) calculate the arithmetic average precipitation across selected weather stations along the Chicago - New York microwave corridor. They define two heavy precipitation intervals: one exceeding half a standard deviation above the mean precipitation in each quarter, and the other exceeding one standard deviation above the mean in each quarter. They assess changes in market quality during these intervals. By averaging precipi-

tation across stations and focusing on time intervals with a minimum average rain frequency, they may ensure that the average weather station in the corridor reports precipitation. This offers a more conservative approach than using station-level data and assuming rain at a single station attenuates microwave signal strength across the entire Chicago – NY corridor.

Thiessen (1911) addresses the challenges in averaging precipitation over large areas. He proposes to use Voronoi diagrams, also known as Thiessen polygons, to estimate each weather station’s area of influence. This method has since been widely used by meteorologists to calculate average precipitation (see, e.g., Nalder and Wein 1998; Ribeiro et al. 2021; and Teegavarapu 2022). We rely on Thiessen’s insight that weighting by polygon areas generated from weather stations as seed points may provide a more robust approximation to whether a large geographic area is experiencing active precipitation, compared to the simple arithmetic mean. Precipitation anywhere along the microwave paths can disrupt the speed of information flow. By aggregating precipitation information using a weighted mean approach, we can more confidently identify disruptions to HFT microwave transmissions.

We capture the average precipitation in the Chicago-New York microwave network corridor using a spherical quadrangle covering the area between the two cities. The northwest and northeast vertices of the spherical quadrangle are located on the data centers of the CME in Aurora, Illinois and the NYSE in Mahwah, New Jersey, respectively. The spherical quadrangle also includes Carteret, New Jersey and Secaucus, New Jersey since these locations house the data centers for the Nasdaq and the Chicago Board of Options Exchange (Cboe), respectively. Based on Laughlin et. al., (2014), we use a 100 km-wide spherical quadrangle (100 km south of each data center) to encompass most microwave transmission networks. We partition the quadrangle using Voronoi diagrams, resulting in 75 polygons, each affiliated with the weather station representing the corresponding area. Precipitation is calculated as a weighted average where each station’s value is weighted by the area it represents. Figure 1 of the main text illustrates the partition. To identify the effect of HFTs on market quality, we focus on 15-minute intervals where the weighted average precipitation along the Chicago –

New York corridor exceeds the thresholds identified by the International Telecommunication Union (ITU) as detrimental to microwave signals. For comparison, we also consider Shkilko and Sokolov’s (2020) endogenous approach, which establishes heavy precipitation thresholds based on the precipitation distribution over quarters within the period studied.

C.1 The arithmetic mean method

Following Shkilko and Sokolov (2020), we define the aggregate precipitation rate P as the simple arithmetic mean of precipitation values from selected reporting stations. During regular trading hours in the NY cluster 9:30am - 4:00pm EST, there are 26 15-minute intervals. Let $t_{k,j}$ denote the j th $j=1, \dots, 26$ 15-minute trading interval on day k . For the weather stations analyzed $i=1, \dots, N$, let $In_{i,t_{k,j}}$ denote the inches of precipitation recorded for station i on interval $t_{k,j}$. Aggregate precipitation rates are calculated using the following expression (1):

$$P_{k,t_{k,j}} = \sum_{i=1}^N \lambda_i \cdot In_{i,t_{k,j}} \quad (1)$$

where λ_i denotes the weight given to each weather station, which equals 1 under the arithmetic mean.

According to FCC records, microwave transmission frequencies used in the Chicago-New York corridor range between 6GHz-22GHz. The ITU (2007) suggests that a precipitation rate of 2.5 mm/hr , or equivalently 0.0246 in/15 min , is sufficient for attenuation impacts on 6GHz signals. Attenuation becomes more pronounced as frequency increases, i.e. a 6GHz signal would be less affected by precipitation than a signal at a higher frequencies. We therefore define the set of time intervals $t_{k,j}$ where precipitation is likely to interfere with microwave transmission, $P(ITU)$, as:

$$P(ITU)_{k,t_{k,j}} := \{x \in P_{k,t_{k,j}} : x > 0.0246in/15min, \text{ else set } 0\} \quad (2)$$

For comparison purposes, we consider the approach by Shkilko and Sokolov (2020) and define two additional attenuation interval sets, $P(1/2StD)$ and $P(1StD)$. The former selects intervals where precipitation exceeds half a standard deviation above the quarterly average, while the latter selects intervals where it exceeds one standard deviation. Let $\bar{P}$ be the quarterly mean precipitation across the different stations over the K days in a quarter,

$$\bar{P} = \frac{1}{KJ} \sum_{k=1}^K \sum_{j=1}^J P_{k,t_{k,j}} \quad (3)$$

with s_P^2 being the variance of the precipitation rate

$$s_P^2 = \frac{1}{KJ-1} \sum_{k=1}^K \sum_{j=1}^J (P_{k,t_{k,j}} - \bar{P})^2 \quad (4)$$

We define $P(1/2StD)$ and $P(1StD)$ as in (5) and (6) respectively,

$$P(1/2StD)_{k,t_{k,j}} := \{x \in P_{k,t_{k,j}} : x > \bar{P} + \frac{1}{2}s_P, \text{ else set } 0\} \quad (5)$$

$$P(1StD)_{k,t_{k,j}} := \{x \in P_{k,t_{k,j}} : x > \bar{P} + s_P, \text{ else set } 0\} \quad (6)$$

While $P(ITU)$ follows scientific guidelines, $P(1/2StD)$ and $P(1StD)$ are based on the precipitation distribution. These measures may either exclude days where precipitation exceeds ITU levels or include days when it falls below those levels. This may result in distorted market quality measures and biased results on the role of HFT on market quality.

C.2 The Thiessen polygon method

Also known as the nearest neighbor method, the Thiessen polygon method assigns precipitation to a geographic point p_i equal to the value registered at the closest weather station. The weights of each weather station are based on the area covered by the points assigned to that station, relative to the total area of the Chicago-New York corridor. Let X be a

metric space and $d(\cdot)$ be the distance metric imposed on X . We define $d(\cdot)$ as the great circle metric.¹ The Thiessen polygon for some fixed point p_i in metric space X is defined as in (7):

$$A_i := \{x \in X : d(x, p_i) \leq d(x, p_j) \forall j \neq i\} \quad (7)$$

The aggregate precipitation rate is calculated as $\sum_{i=1}^N \lambda_i \cdot In_{i,t_{k,j}} = \frac{\sum_{i=1}^N \text{Area of } A_i \times In_{i,t_{k,j}}}{\sum_{i=1}^N \text{Area of } A_i} =: P_{k,t_{k,j}}^*$, where λ_i equals the ratio between the area covered by the polygon A_i and the total area of the collection of Thiessen polygons $\{A_i\}_{i=1}^N$. To define the latter, we use a spherical quadrangle² of 100km width since it reasonably covers the Chicago and New York microwave transmission corridor.

The collection of Thiessen polygons $\{A_i\}_{i=1}^N$ inscribed on the spherical quadrangle are shown in Figure 1 in the main text. The points on the diagram represent the locations of the weather stations used to generate the polygons. The northwestern and northeastern vertices of the spherical quadrangle represent the CME and NY cluster data centers, respectively, following the Laughlin et al. (2014) appendix Figure A1. Figure 1 in the main text displays the location of Mahwah, Secaucus and Carteret which house the matching engines of NYSE, NASDAQ, and Cboe exchanges, the most active exchanges over the period of analysis.

The gray points in Figure 1 represent weather stations that closed during the study period (see Tables C.1, and C.2 for details on selected weather stations and those that closed, respectively). When a station closes, we remove that station from the aggregate precipitation measure and do not assign its corresponding area to any other station. This prevents the need to generate a new Thiessen collection based on the remaining weather stations, which could yield misleading results when statistically compared to the arithmetic

¹For any two points in X , say p_1 and p_2 then $d(p_1 = (x, a), p_2 = (y, b)) = r \cos^{-1}(\cos(a)\cos(b)\cos(x - y) - \sin(a)\sin(b))$. For example, the point p_1 represents a coordinate composed of longitude in the x-coordinate and latitude in the y-coordinate, where r is the radius of planet Earth.

²A recent example of the use of an ellipsoid instead of a sphere for the creation of Thiessen polygons is found in Kastrisios and Tsoulos (2018). The upper bound error of the use of a sphere over an ellipsoid is 0.05% which for this setting is reasonably tolerable. Implementation of the Thiessen polygon method is made possible by the geopandas Python library.

mean method in our multi-year statistical analysis. Generating new Thiessen collections would be appropriate if considering single year analyses. As a robustness check, we remove the closing weather stations, that is we use weather stations that are consistently in operation throughout our sample, and generate new weights shown in Figure C.1 with precipitation summary statistics shown in Table C.3. Summary statistics computed are similar to those in the main text found in Table 2.

Similar to $P(ITU)$, $P(1/2StD)$ and $P(1StD)$ which are based on the mean method, we create $P(ITU)^*$, $P(1/2StD)^*$ and $P(1StD)^*$ based on the Thiessen method. In the next section we detail the procedure to calculate the coordinates of the spherical quadrangle.

C.3 Spherical quadrangle calculator

The Chicago-New York microwave network corridor average precipitation can be captured by a spherical quadrangle partition of the Voronoi diagram covering the majority of the microwave transmission networks. Using the Laughlin et. al., (2014) diagram it is reasonable to consider a 100km width spherical quadrangle (100km southward from each data center). To create the partition we use Python with the geopandas library. The partition yields 75 polygons each affiliated with the weather station representing that area. We calculate the areas in kilometers squared using the Lambert Azimuthal Equal-Area spherical projection for the United States—this projection maps the sphere to a disk and computes the areas of the polygons.³ The area of the 100km spherical quadrangle computed using this method is $125,589.0km^2$ while the sum of the partitioned polygons is $125,601.2km^2$ which represents an error of 0.00971%. When performing this operation no weather stations in Michigan are available because the residual areas from polygons generated in Michigan do not integrate the partition. We proceed to create the arithmetic mean and Thiessen polygon precipitation variables.

With the geolocation information of the two data centers from CME Aurora, IL and

³See details at <https://spatialreference.org/ref/epsg/us-national-atlas-equal-area/>

NYSE Mawah, NJ we calculate the normal vectors associated to the points (note that latitude = a and longitude = b should be in radians) (Stewart, J. 2016).

$$\text{Normal vector} = (\cos(a)\cos(b), \cos(a)\sin(b), \sin(a))$$

We then obtain the cross product of the two normal vectors (which is a vector (θ, ϕ, r) in spherical coordinates). We normalize the cross product (i.e., divide by its magnitude). Using the Cartesian coordinates of CME and NYSE we add to each xyz coordinates the distance desired (-100km in this case since the direction is southward, or w) times the normalized cross product direction θ , ϕ , and r . This yields the southward vertices desired. Using the great circle distance length this can be readily employed to find the lengths of the arcs from Chicago to NY (or l).

C.4 Precipitation analysis

The National Centers for Environmental Information (NCEI) (previously known as the National Climatic Data Center (NCDC) which dissolved in 2015) are operated in part by an office of the National Oceanographic and Atmospheric Administration (NOAA). The NCDC data set 3260 (DSI-3260) also known as the 15 Minute Precipitation Data is available from 1971 through 2013. We present an example of the data set in Table C.4. Since geo-location station data is not found for this period, we use the geo-location station data in NCEI's 2014 and onwards Hourly Precipitation Data set (HPDv2 beta version). We focus on 2011-2017 data from the DSI-3260⁴ and the HPDv2 beta version⁵ that provides 15-minute precipitation intervals. We impute 0 to precipitation values above 3 inches, representing a conservative estimate of extreme precipitation.

We intersect the HDPv2 station data with the the 2011-2013 DSI-3260 station data to

⁴DSI-3260 File transfer protocol available through ftp://ftp.ncdc.noaa.gov/pub/data/15min_precip-3260/, documentation is found within the server.

⁵The data set is retrieved through this public link <https://www.ncei.noaa.gov/pub/data/hpd/auto/v2/beta/15min/> along with documentation information <https://www.ncei.noaa.gov/pub/data/hpd/auto/v2/beta/15min/readme.csv.txt>

obtain a list of weather stations operating in Illinois, Indiana, Michigan, Ohio, Pennsylvania, and New Jersey for 2011 and onwards. When performing this operation, no weather stations in New Jersey are available. These geolocations are used to create the spherical Thiessen Polygon (Voronoi diagram naming scheme for this mathematical object is used in the Python documentation) using Python programming language library SciPy spatial.

The network of affiliated weather stations reports precipitation in inches using Fisher-Porter (F&P) gauges with accuracy up to one-hundredths of an inch. The HPDv2 beta version readme file states precipitation data is in hundredths of an inch. However, if this is the case the values are one magnitude too small (see footnote 5 of this appendix for precipitation data set documentation), suggesting that they are actually expressed in tenths of an inch (i.e., divide by 10 instead of 100, dividing by 100 results in no precipitation days while not dividing the value results in a disproportional amount of precipitation days). We account for this discrepancy in our data cleaning procedure. Table C.5 shows an HPDv2 beta version example of precipitation data for a particular weather station. Information about weather station location along with ID, latitude, longitude, Thiessen polygon area, state, and name are found in Table C.1. Table C.2 shows this same information but for weather stations that close during our period of analysis.

Precipitation statistics are computed for standard trading days (9:30-16:00 EST), during which exchanges in the New York cluster are typically operational. As a robustness check we verify that the summary statistics align with those computed by Shkilko and Sokolov (2020) for the 2011-2012 period using 75 weather stations instead of 54 used by these authors. Table C.6 Panel’s A and B present a comparison of precipitation summary statistics between the arithmetic mean and Thiessen polygon methods. Panel A removes all null values, while Panel B removes all null values along with values greater than 3in/15min. Panel C provides precipitation results derived by Shkilko and Sokolov (2020). We note that the values for the computed statistics in both Panel A and B are the same. Panel A and B values are also similar to Panel C. Therefore we are confident in that the methodology employed to analyze

the precipitation data is adequate.

Table C.7 presents the total amount of observations when considering each weather station each year. We present the amount of null values and those greater than 3in/15min. Here is a helpful breakdown, consider the year 2011 for example. When aggregating the observations per weather stations by each 15 minute interval they amount to 6804. There are 75 weather stations in our sample, thus there are a total of $75 \cdot 6804 = 510300$ observations. Since null values and those greater than 3in/15min is 7210 then $(7210/510300) \cdot 100 = 1.41\%$ represents the percentage of removed observations. We note that there is a loss of around 1% of precipitation observations per year. However, we are confident that this small amount does not influence our results.

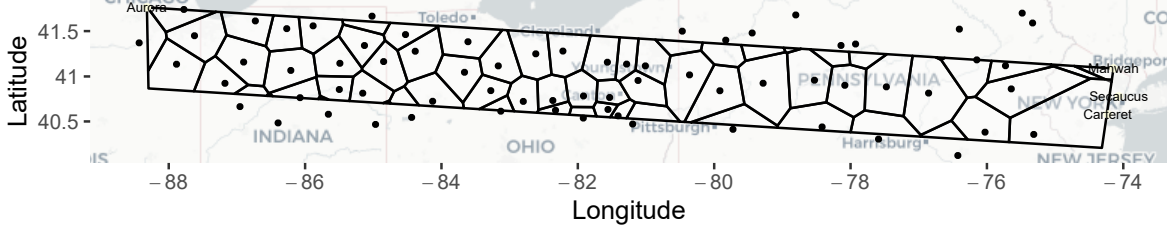


Fig. C.1. Thiessen polygons inscribed on a spherical quadrangle

Note: This figure illustrates the partition of the area covering the microwave network corridor between Chicago and New York. The area is represented by a spherical quadrangle. The northwest and northeast vertices of the spherical quadrangle are located on the data centers of the Chicago Mercantile Exchange in Aurora, Illinois and the New York Stock Exchange in Mahwah, New Jersey, respectively. The spherical quadrangle includes Carteret, New Jersey and Secaucus, New Jersey since these locations house the data centers for the National Association of Securities Dealers Automated Quotations and the Chicago Board Options Exchange, respectively. Based on Laughlin et. al., (2014), we use a 100 km-wide spherical quadrangle (100 km south of each data center) to encompass most microwave transmission networks. The polygon vertices are generated using US National Atlas Equal Area (EPSG:2163) and forced to plot on World Geodetic System (EPSG:4326). We partition the quadrangle using Thiessen polygons, resulting in 70 polygons, each affiliated with the weather station representing the corresponding area. Precipitation is calculated as a weighted average where each station's value is weighted by the area it represents. The black points represent the locations of the weather stations used to generate the Thiessen Polygons. This figure may be compared to Figure 1 in the main text which accommodates closing stations.

Table C.1. Weather station information

Station ID	Latitude	Longitude	Thiessen Area	State	Name
USC00111577	41.7372	-87.7775	1288.10339	IL	CHICAGO MIDWAY AP 3SW
USC00112011	41.4492	-87.6221	2350.05152	IL	CRETE
USC00114603	41.1351	-87.8856	2808.51883	IL	KANKAKEE METRO WASTEWATER
USC00115825	41.3708	-88.4336	1066.35141	IL	MORRIS 1 NW
USC00116819	40.7569	-88.1827	484.96205	IL	PIPER CITY
USC00120200	41.6639	-85.0183	463.49030	IN	ANGOLA
USC00120830	40.8143	-85.1546	1556.49663	IN	BLUFFTON 6N
USC00121147	40.4835	-86.3961	15.40830	IN	BURLINGTON
USC00121417	40.6653	-86.9550	398.27662	IN	CHALMERS 5 W
USC00121739	41.1452	-85.4898	2075.67434	IN	COLUMBIA CITY
USC00123206	41.3415	-85.1292	1473.14102	IN	GARRETT
USC00123418	41.5575	-85.8825	1525.62378	IN	GOSHEN 3SW
USC00124181	40.8555	-85.4980	1253.51576	IN	HUNTINGTON
USC00124730	41.6474	-85.4121	754.59721	IN	LAGRANGE
USC00124782	41.5269	-86.2692	1696.81058	IN	LAKEVILLE
USC00124837	41.6116	-86.7297	1119.61258	IN	LAPORTE
USC00125337	40.5800	-85.6586	28.89089	IN	MARION 2 N
USC00125535	41.1588	-86.9013	2529.62728	IN	MEDARYVILLE 5 N
USC00126864	40.7646	-86.0740	1017.97282	IN	PERU
USC00127069	40.4667	-84.9667	17.54290	IN	PORTLAND WATER WORKS
USC00127298	40.9239	-87.1754	1938.55990	IN	RENSSELAER
USC00127482	41.0665	-86.2096	2767.68145	IN	ROCHESTER
USC00128992	41.5436	-87.0319	1743.19780	IN	VALPARAISO 5NNE
USC00129724	41.1645	-84.8482	2092.23105	IN	WOODBURN 3N
USC00330059	41.1573	-81.5664	1773.31036	OH	AKRON WPCS
USC00330107	40.9551	-81.1167	1785.43133	OH	ALLIANCE 3 NNW
USC00330493	40.6335	-81.5577	441.50727	OH	BEACH CITY LAKE
USC00330639	41.1169	-81.0094	1815.87478	OH	BERLIN LAKE
USC00330862	41.3831	-83.6111	2386.48422	OH	BOWLING GREEN WWTP
USC00331042	41.4619	-84.5272	1343.91209	OH	BRYAN 2 SE
USC00331466	40.7333	-82.3667	1339.78431	OH	CHARLES MILL LAKE
USC00331541	41.0517	-81.9361	1430.30444	OH	CHIPPEWA LAKE
USC00332098	41.2783	-84.3847	2467.92922	OH	DEFIANCE
USC00332272	40.5622	-81.4092	433.15818	OH	DOVER DAM
USC00332791	41.0461	-83.6622	2268.20548	OH	FINDLAY WPCC
USC00333021	40.7231	-82.7999	1503.13580	OH	GALION WATER WORKS
USC00334473	40.4709	-81.1955	1079.93444	OH	LEESVILLE LAKE
USC00334551	40.7247	-84.1294	2062.56865	OH	LIMA WWTP
USC00334942	40.6142	-83.1300	444.42857	OH	MARION 2 N
USC00334992	40.7667	-81.5333	1041.78379	OH	MASSILLON

Station ID	Latitude	Longitude	Thiessen Area	State	Name
USC00335297	40.5375	-81.9197	342.49622	OH	MILLERSBURG
USC00336123	41.2500	-82.6167	2546.85036	OH	NORWALK HIWAY DEPT
USC00336196	41.2802	-82.2187	1892.51190	OH	OBERLIN
USC00336702	40.6245	-82.3271	440.60286	OH	PLEASANT HILL LAKE
USC00336949	41.1366	-81.2844	1202.55481	OH	RAVENNA 2 S
USC00337383	40.5449	-84.4374	528.14194	OH	ST. MARYS 3 W
USC00338313	41.1167	-83.1667	2262.82312	OH	TIFFIN
USC00338539	40.8428	-83.2767	1602.50321	OH	UPPER SANDUSKY WTR WK
USC00339312	40.7833	-81.9167	1064.75402	OH	WOOSTER EXP STATION
USC00360147	41.3579	-77.9264	654.17078	PA	ALVIN R BUSH DAM
USC00360560	40.8620	-75.6429	4796.63887	PA	BELTZVILLE DAM
USC00360725	40.3027	-77.5894	2012.83990	PA	BLAIN 5SW
USC00360785	40.3803	-76.0275	2996.52679	PA	BLUE MARSH LAKE
USC00360821	40.1597	-79.0287	474.56613	PA	BOSWELL
USC00361139	40.8415	-79.9163	3175.01325	PA	BUTLER 2 SW
USC00361961	40.9583	-78.5294	3438.49913	PA	CURWENSVILLE LAKE
USC00362245	41.3421	-78.1401	684.69389	PA	DRIFTWOOD
USC00362323	41.5216	-76.4043	16.75366	PA	DUSHORE
USC00363018	41.1184	-75.7277	1195.94005	PA	FRANCIS E WALTER DAM
USC00363028	41.4004	-79.8305	1321.80255	PA	FRANKLIN
USC00364001	40.4388	-78.4173	2390.18335	PA	HOLLIDAYSBURG 2NW
USC00364325	41.4992	-80.4681	769.26328	PA	JAMESTOWN 2 NW
USC00364432	41.6770	-78.8036	75.42884	PA	KANE 1NNE
USC00364778	40.1190	-76.4265	731.98729	PA	LANDISVILLE 2 NW
USC00365790	40.8847	-77.4750	3836.89871	PA	MILLHEIM
USC00366111	40.4120	-79.7245	1477.45163	PA	MURRYSVILLE 2 SW
USC00366233	41.0172	-80.3615	3636.93842	PA	NEW CASTLE 1 N
USC00366916	40.9017	-78.0842	2672.88560	PA	PHILIPSBURG 8 E
USC00367186	41.5890	-75.3303	393.71225	PA	PROMPTON DAM
USC00367229	40.9248	-79.2825	4345.08462	PA	PUTNEYVILLE 2 SE DAM
USC00367931	40.8153	-76.8556	5446.39131	PA	SELINSGROVE CAA AIRPORT
USC00367938	40.3552	-75.3131	7963.07892	PA	SELLERSVILLE
USC00368057	41.1831	-76.1488	1998.59083	PA	SHICKSHINNY 3 N
USC00368491	41.6972	-75.4827	436.63097	PA	STILLWATER DAM
USC00368873	41.4792	-79.4432	692.36930	PA	TIONESTA 2 SE LAKE

Note: This table reports the weather stations used in our analysis, along with their ID, latitude, longitude, Thiessen polygon area, state, and name.

Table C.2. Closing weather station's information

Station ID	Closing Year	Latitude	Longitude	Thiessen Area	State	Name
USC00116819	2013	40.7569	-88.1827	484.9621	IL	PIPER CITY
USC00128992	2014	41.5436	-87.0319	1743.1978	IN	VALPARAISO 5NNE
USC00124730	2015	41.6474	-85.4121	754.5972	IN	LAGRANGE
USC00331541	2015	41.0517	-81.9361	1430.3044	OH	CHIPPEWA LAKE
USC00360821	2017	40.1597	-79.0287	474.5661	PA	BOSWELL

Note: This table reports the weather stations that closed during the period of analysis, along with the year when they closed, their ID, latitude, longitude, Thiessen polygon area, state, and name.

Table C.3. Aggregate precipitation rate statistics in inches per 15-minute interval

	Min	Mean	Median	St. Dev	Max	% of Int w. Precip
Panel A: Arithmetic Mean Method						
P	0	0.00337	0	0.0083	0.22857	34.27135
P(1/2 StD)	0	0.00265	0	0.00834	0.22857	13.47929
P(1 StD)	0	0.00233	0	0.0082	0.22857	10.32452
P(ITU)	0	0.00122	0	0.00729	0.22857	3.15057
Panel B: Thiessen Polygon Method						
P*	0	0.0035	0	0.00896	0.2536	34.27135
P(1/2 StD)*	0	0.00276	0	0.00898	0.2536	12.97453
P(1 StD)*	0	0.00237	0	0.00882	0.2536	9.34233
P(ITU)*	0	0.00143	0	0.00805	0.2536	3.62168

Panel C	P(1/2 StD)	P(1 StD)	P(ITU)	P(1/2 StD)*	P(1 StD)*	P(ITU)*
P(1/2 StD)		p < 0.01	p < 0.01	p = 0.072	p < 0.01	p < 0.01
P(1 StD)			p < 0.01	p < 0.01	p < 0.01	p < 0.01
P(ITU)				p < 0.01	p < 0.01	p < 0.01
P(1/2 StD)*					p < 0.01	p < 0.01
P(1 StD)*						p < 0.01
P(ITU)*						

Note: This table presents summary statistics on aggregate precipitation and percent of intervals with precipitation (% of Int w. Precip) on the Chicago – New York corridor. Summary statistics are expressed in inches for each 15-minute interval during market trading hours (9:30 – 4:00 pm EST) for the period from 2011 to 2017. Total number of intervals is 47,547. We consider both the arithmetic mean (Panel A) and the Thiessen polygon method (Panel B). We distinguish between the two methods through the variable names; an asterisk after the variable name indicates Thiessen polygon-based measures. For each of these methods we consider three different thresholds to select heavy precipitation intervals identified based on the actual rain distribution (P(1/2 StD) and P(1 StD)) and on the ITU guidelines (P(ITU)). P denotes all precipitation intervals before filtering the heavy precipitation intervals using thresholds. We test whether the computed aggregate precipitation measures differ using the Wilcoxon signed-rank test. The Wilcoxon signed-rank test for each pair of measurements, namely, P(1/2 StD)* vs. P(1/2 StD), P(1 StD)* vs. P(1 StD), and P(ITU)* vs. P(ITU) with null hypothesis of equal medians is rejected. The p-values for these tests are found in Panel C. Jarque-Bera tests for normality are rejected in all instances.

Table C.4. An example of 15-minute precipitation data DSI-3260

State	Coop. Index	Units	Year	Month	Day	Time	Value
11	0082	HT	2011	12	0001	0015	00000
11	0082	HT	2011	12	0003	0345	00010
11	0082	HT	2011	12	0013	1930	00010
11	0082	HT	2011	12	0014	0015	00010
11	0082	HT	2011	12	0021	0445	00010
11	0082	HT	2011	12	0030	0645	00010
11	0082	HT	2011	12	0001	0015	00484
11	0082	HT	2011	12	0003	0345	00485
11	0082	HT	2011	12	0004	0115	00490
11	0082	HT	2011	12	0006	0915	00491

Note: This table provides a subset of 15-minute precipitation data from the DSI-3260 (NCEI) for the 2011-2013 period. The State column identifies the state where the observation was recorded, using numeric codes (e.g., 11 for Illinois) as defined in the DSI-3260 state code table. The Coop. Index column specifies the 4-digit Cooperative Network Index number assigned by the National Climatic Data Center (NCDC) to uniquely identify the station within the cooperative observer network. The Units column denotes the measurement units for precipitation, with HT indicating values are recorded in tenths of inches and HI (not part of the example below) in hundredths of an inch. The Year, Month, and Day columns provide the calendar date for each observation. The Time column records the end of the 15-minute interval during which the precipitation measurement was taken. For example, 0015 represents data for the period from 0001 to 0015 local standard time. The Value column contains the precipitation amount for the interval, expressed in the specified units. For instance, a value of 00010 measured in HT units indicates 0.10 inches of precipitation.

Table C.5. An example of 15-minute precipitation data HPDv2 beta version

Station ID	Latitude	Longitude	Elevation	Date	0000 Value	...	2345 Value
USC00111577	41.7372	-87.7775	189	2014-01-01	0	...	0
USC00111577	41.7372	-87.7775	189	2014-01-02	0	...	0
USC00111577	41.7372	-87.7775	189	2014-01-03	0	...	0
USC00111577	41.7372	-87.7775	189	2014-01-04	0	...	1
USC00111577	41.7372	-87.7775	189	2014-01-05	0	...	0
USC00111577	41.7372	-87.7775	189	2014-01-06	0	...	0
USC00111577	41.7372	-87.7775	189	2014-01-07	0	...	0
USC00111577	41.7372	-87.7775	189	2014-01-08	0	...	0
USC00111577	41.7372	-87.7775	189	2014-01-09	0	...	0
USC00111577	41.7372	-87.7775	189	2014-01-10	0	...	0

Note: This table provides a subset of 15-minute precipitation data from the Hourly Precipitation Data version 2 (HPDv2) beta version (NCEI) for the 2014-2017 period. The Station column identifies the weather station code where the last four digits are the Cooperative Network Index digits. The Latitude and Longitude columns denote the station's geographic coordinates. The Elevation column denotes the weather station's feet above sea level. The Date column provides the calendar date for each observation. The Value columns records the end of the 15-minute interval during which the precipitation measurement was taken. For example, 0015 represents data for the period from 0001 to 0015 in local standard time. To our understanding, these values are expressed in tenths of an inch. For example on 2014-01-04 during the 2345 15 minute interval starting at 2331 a total of 0.1 inches of precipitation were measured.

Table C.6. Aggregate precipitation summary statistics comparison

Panel A	Thiessen	Arithmetic
Mean	0.002	0.002
Standard Deviation	0.005	0.004
Median	0.000	0.000
Panel B	Thiessen	Arithmetic
Mean	0.002	0.002
Standard Deviation	0.005	0.004
Median	0.000	0.000
Panel C	Thiessen	Arithmetic
Mean		0.002
Standard Deviation		0.005
Median		0.001

Note: This table presents summary statistics for the precipitation measures during the 2011-2012 sample period. We consider the constructed aggregate precipitation methods, Thiessen (P*) and the Arithmetic Mean (P). Panel A contains statistics for aggregate precipitation, excluding all null values 2011-2012. Panel B removes all null values and those greater than 3in/15min. The summary statistics in Panel B correspond to the data used in the main analysis. Panel C provides precipitation results derived by Shkilko and Sokolov (2020) found in Table 1 of their research article.

Table C.7. Analysis of null values and those greater than 3in/15min per year

Year	Observations	Null Values	Null Values & >3	% Null Values	% Null Values & >3
2011	510300	3905	7210	0.76%	1.41%
2012	506250	3479	5726	0.68%	1.13%
2013	510300	1375	4158	0.27%	0.81%
2014	483084	1466	5668	0.30%	1.14%
2015	483084	963	4314	0.20%	0.89%
2016	483084	1253	4237	0.26%	0.88%
2017	474390	1385	4969	0.29%	1.05%

Note: This table presents the total amount of observations when considering each weather station by year. We present the number of null values and those greater than 3in/15min. For example: consider the year 2011. When aggregating the observations per weather stations for each 15 minute interval they amount to 6804. There are 75 weather stations in our sample, thus there are a total of $75 \cdot 6804 = 510300$ observations. Since null values and those greater than 3in/15min are 7210 then $(7210/510300) \cdot 100 = 1.41\%$ represents the percentage of removed observations. We note that there is a loss of around 1% of precipitation observations per year. However, we are confident that this small amount does not influence our results.

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Appendix D

This appendix examines microwave transmissions from HFTs involved in spatial latency arbitrage. The analysis focuses on observing potential trade delays resulting from HFTs switching between microwave networks and fiber optic cable transmissions due to precipitation. Empirically, microwave transmissions typically take between 4.5 to 5 milliseconds, whereas fiber optic transmissions require approximately 6.5 to 8 milliseconds, resulting in a difference of 2 to 3 milliseconds. If HFTs fail to automatically switch to fiber optics, information transmission may still occur, albeit with potential loss; during precipitation-induced fading, the use of more robust error control, buffering, or recovery mechanisms can increase tail latency and, in some implementations, generate millisecond-scale delays (Sklar 2001). We attempt to detect differences in trading behavior during precipitation vs. non-precipitation periods within a single trading day.

We study the responses of energy ETFs traded in the NY cluster to trades in the CME futures market. To ensure that trades in the former respond to information from the latter, we first identify standalone futures signals — defined as trades in the futures market not preceded by other trades during the previous 10 milliseconds — and second, trades that represent a change in price direction per the Lee and Ready (1991) algorithm. This is done in similar fashion to Rösch (2021). By requiring either time separation or direction changes, we aim to capture genuine HFT activity while filtering out noise from regular trading patterns, thus helping to distinguish deliberate HFT strategies from regular market activity or order splitting. We investigate trade activity in the NY cluster securities within 5 to 10 milliseconds of these selected futures market signals. Given current technology speeds, the minimum time to observe HFT responses is 4.5 milliseconds. Further, we cannot be certain that trade activity beyond 10 milliseconds originates from the HFTs focused on spatial latency arbitrage.

To conduct the analysis we use market depth data from the CME that includes trade and quote information time stamped to the millisecond. We select the months of June 2013 and

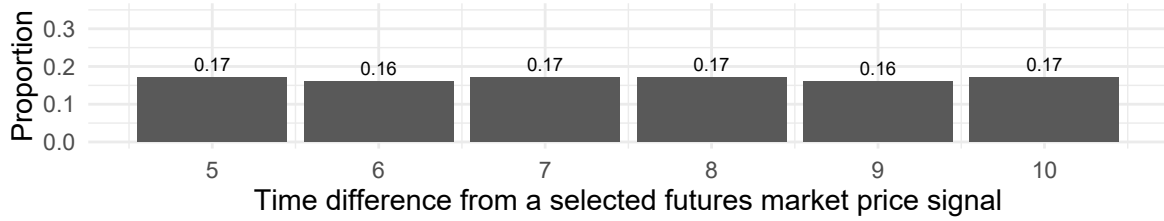
March 2016 due to a higher incidence of disruptive precipitation periods during Mondays, Thursdays, and Fridays which minimizes concerns about possible distortions caused by the American Petroleum Institute and Energy Information Administration announcements on Tuesdays and Wednesdays. Table D.1 provides a summary of the dates analyzed by price pair (first and third columns), disruption dates analyzed (second column) along with the aggregate number of 15 minute precipitation disruption intervals for these dates (fourth column). To select precipitation disruptions we use the ITU-Thiessen polygon approach discussed in appendix C.

Identifying HFT is challenging since their behavior might manifest in various ways depending on the intensity of their activities. In Figure D.1, we develop two examples that capture two of the possible expected behaviors. Example 1 details that during non-disrupted periods, trading activity is uniformly distributed across millisecond bins, suggesting HFT participation is masked by overall market trading volume; during precipitation disruptions however, there is a concentrated spike in trading at intermediate milliseconds (ms 7-8), consistent with expected delays from microwave to fiber-optic switching. Example 2 presents an alternative pattern: during non-disrupted periods, HFTs exhibit more aggressive trading concentrated in early millisecond bins (5-6ms); during disruptions, trading spikes appear in later milliseconds (ms 9-10), suggesting HFT trade algorithms are delayed by the technology switch or due to implementing microwave signal correction algorithms due to signal attenuation. Both examples demonstrate how precipitation disruptions may affect the timing and pattern of HFT activity, either through simple transmission delays or through more complex responses.

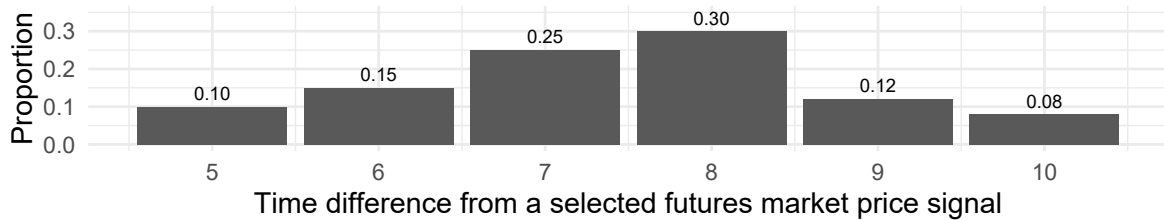
Figures D.2 through D.4 show the empirical results of this procedure for the energy futures and ETF pairs. Figure D.2 shows the pairs (NG-UNG) and (CL-USO). The CL-USO pair found in the bottom panel of figure D.2 is found in the main text as figure 2, and to facilitate comparison with the examples provided above it is included. The (NG-UNG) plots in this figure are developed using June 2013 data and show a similar pattern to Example 2 except

that the 5ms bin trade is fairly large during precipitation. This can be attributed to trades not originating from HFTs that just happen to occur at that moment or microwave signals that manage to reach NY without information loss. The (CL-USO) plot presents a consistent pattern with Example 1. Figure D.3 shows results for the (CL-UCO) and (CL-OIL) pairs. The (CL-UCO) plot is similar to Figure D.1's (NG-UNG) pattern for the 5ms bin, for which the same explanation applies. For (CL-OIL) pair the similarities with the example patterns are clearer than the other pairs involving CL. This is a somewhat surprising result since the OIL ETF is the smallest in terms of market capitalization when compared to the USO and UCO ETFs.

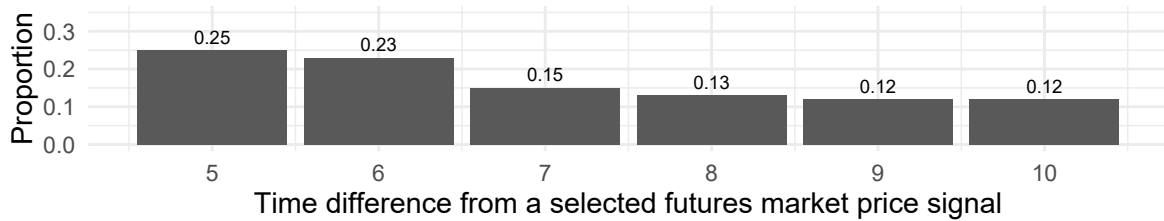
Trading behavior during non disruptive intervals (example 1)



Trading behavior during precipitation disruption intervals (example 1)



Trading behavior during non disruptive intervals (example 2)



Trading behavior during precipitation disruption intervals (example 2)

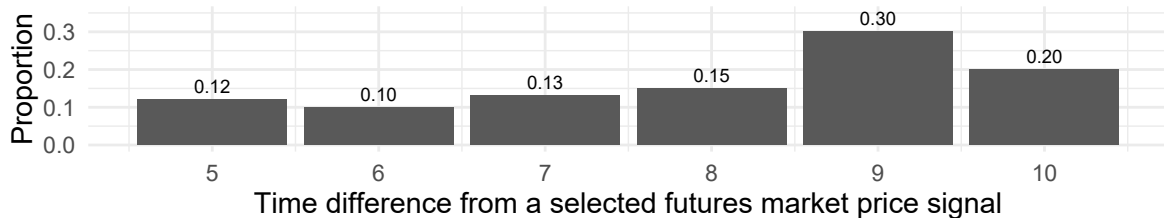
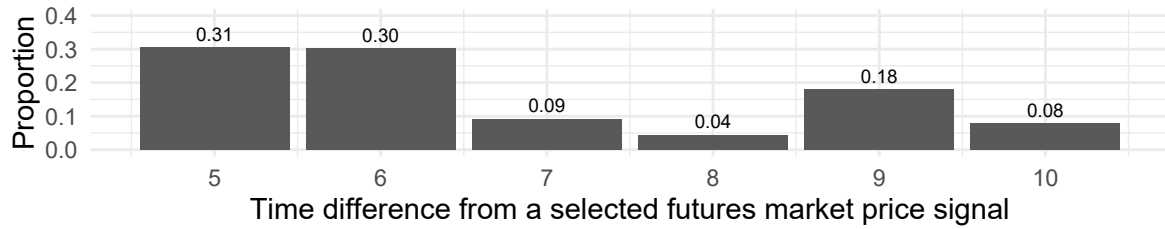


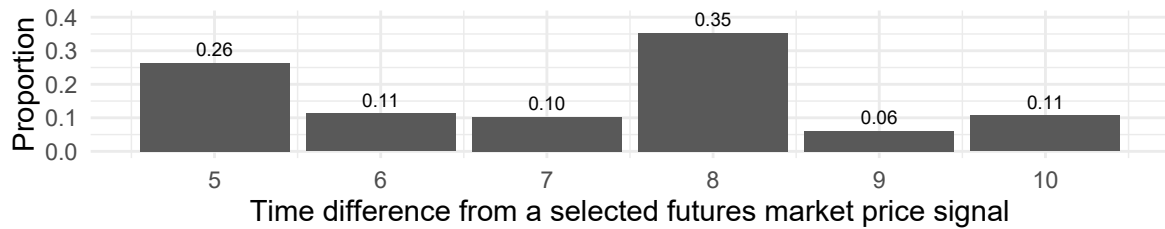
Fig. D.1. Examples of HFT trading patterns following selected price signals.

Note: This figure presents two possible patterns of trading behavior in the ETF market following selected price signals in the futures market. In Example 1 during non-disrupted periods, trading activity is uniformly distributed across millisecond bins, suggesting HFT participation is masked by overall market trading volume; during precipitation disruptions, there is a concentrated spike in trading at intermediate milliseconds (ms 7–8), consistent with expected delays from microwave to fiber-optic switching. Example 2 presents an alternative pattern: during non-disrupted periods, HFTs exhibit more aggressive trading concentrated in early millisecond bins (5–6 ms); during disruptions, trading spikes appear in later milliseconds (ms 9–10), suggesting HFT trade algorithms are delayed by precipitation.

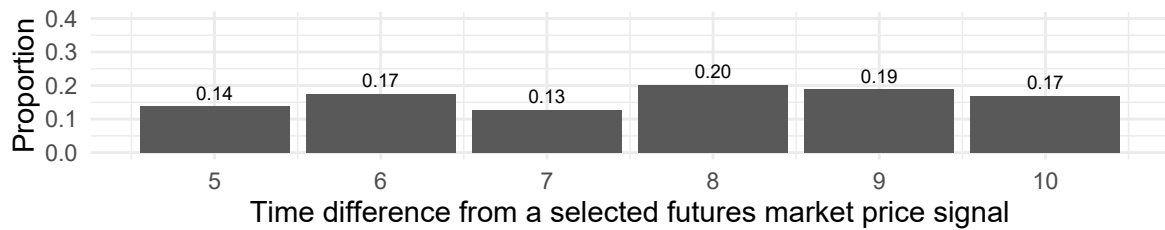
Trading behavior during non disruptive intervals (NG UNG)



Trading behavior during precipitation disruption intervals (NG UNG)



Trading behavior during non disruptive intervals (CL USO)



Trading behavior during precipitation disruption intervals (CL USO)

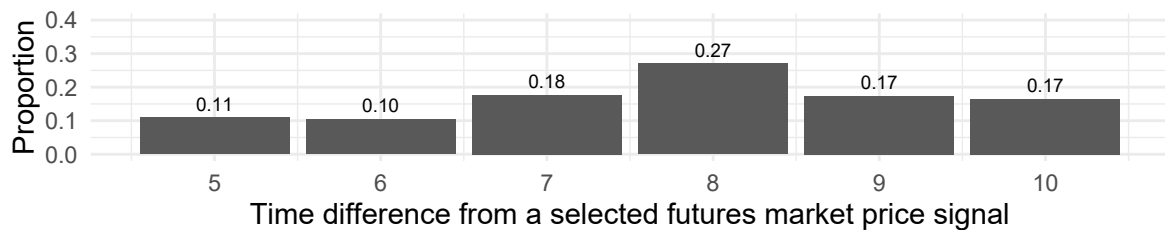
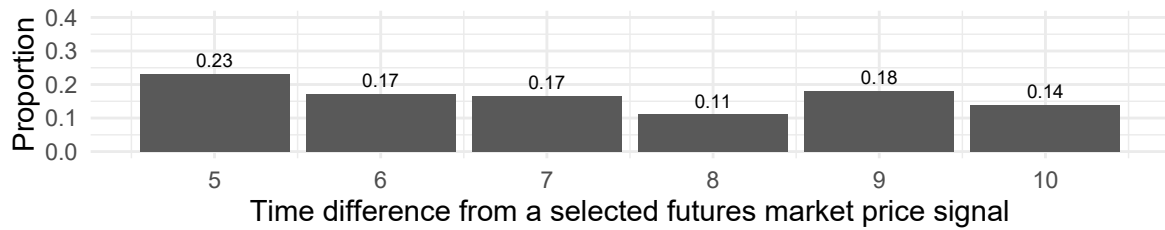


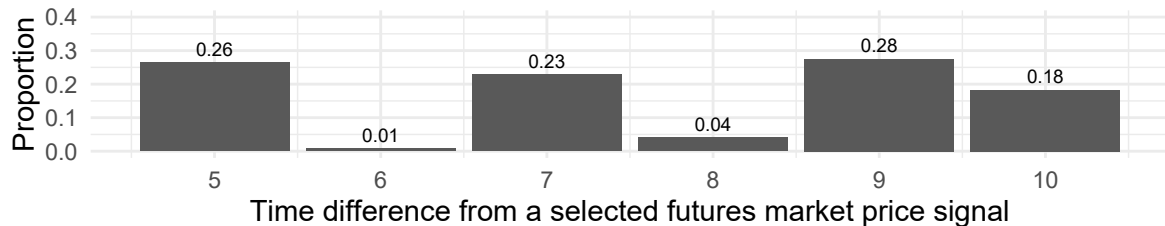
Fig. D.2. UNG and USO trades following NG and CL selected price signals

Note: This figure presents the United States Natural Gas Fund LP (UNG) and United States Oil Fund LP (USO) ETF reactions to selected price signals originating from the Henry Hub Natural Gas (NG) and the West Texas Intermediate Crude Oil (CL) futures. The month of June 2013 is selected for (NG-UNG) due to a higher incidence of disruptive precipitation periods during Mondays, Thursdays, and Fridays which removes concerns about distortions created by the American Petroleum Institute and Energy Information Administration announcements. The months of June 2013 and March 2016 for (CL-USO) are chosen for the same reason.

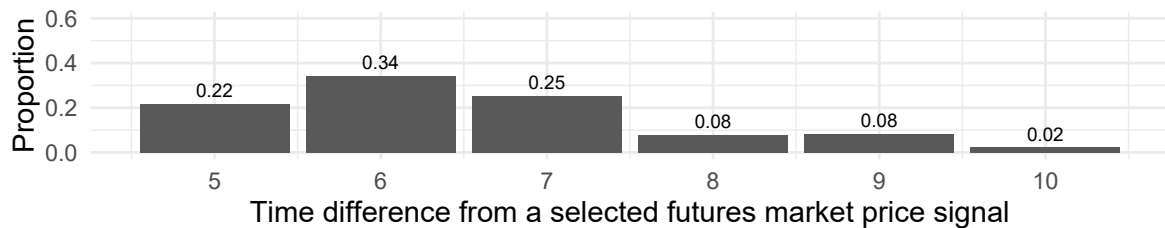
Trading behavior during non disruptive intervals (CL UCO)



Trading behavior during precipitation disruption intervals (CL UCO)



Trading behavior during non disruptive intervals (CL OIL)



Trading behavior during precipitation disruption intervals (CL OIL)

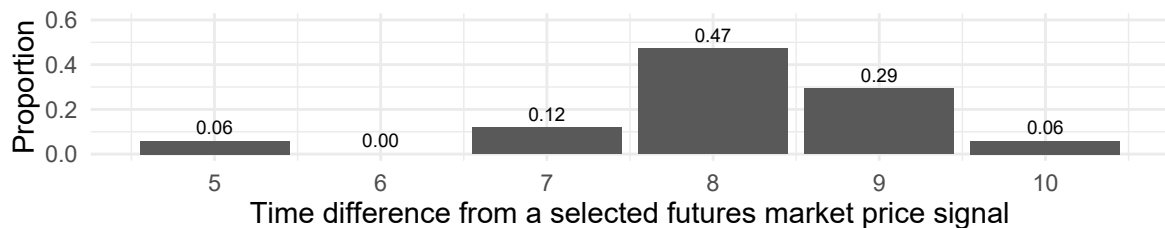


Fig. D.3. UCO and OIL trades following CL selected price signals

Note: This figure presents the ProShares Ultra Bloomberg Crude Oil (UCO) and iPath Pure Beta Crude Oil ETN (OIL) ETF reactions to selected price signals originating from the West Texas Intermediate Crude Oil (CL) futures. The months of June 2013 and March 2016 are selected due to a higher incidence of disruptive precipitation periods during Mondays, Thursdays, and Fridays which removes concerns about distortions created by the American Petroleum Institute and Energy Information Administration announcements.

Table D.1. Dates analyzed for selected price series pairs

Period	Disruption Dates	Pairs	# Disruptions
2013-06	2013-06-06, 2013-06-07, 2013-06-17, 2013-06-20, 2013-06-21, 2013-06-27, 2013-06-28	NG-UNG, CL-USO, CL-UCO, CL-OIL	19
2016-03	2016-03-04, 2016-03-10, 2016-03-24, 2016-03-31	CL-USO, CL-UCO, CL-OIL	8

Note: This table provides a summary of the price series pairs and disruption dates analyzed along with the aggregate number of 15 minute precipitation disruption intervals for the selected dates. For this exercise we select precipitation disruptions using the ITU-Thiessen polygon threshold approach discussed in appendix C.

Table D.2. An example of CME market by price (MBP) data

Date	Time	Type	Side	Price	Size	Depth	Order	Message Seq #	Contract
2013-06-06	22:00:00.323	Submission	Bid	\$4.102	1	1		5079402	NGN4
2013-06-06	22:00:00.323	Submission	Bid	\$4.096	1	2		5079402	NGN4
2013-06-06	22:00:00.323	Submission	Ask	\$4.254	1	1		5079402	NGN4
2013-06-06	22:00:00.323	Submission	Ask	\$4.255	1	2		5079402	NGN4
2013-06-06	22:00:00.431	Submission	Bid	\$4.050	16	2	1	5079421	NGN4
2013-06-06	22:00:00.431	Submission	Ask	\$4.250	16	1	1	5079421	NGN4
2013-06-06	22:00:00.431	Submission	Ask	\$4.235	18	1		5079421	NGN4
2013-06-06	22:00:00.431	Submission	Ask	\$4.236	6	2		5079421	NGN4
2013-06-06	22:00:00.440	Submission	Bid	\$4.000	16	3	1	5079468	NGN4
2013-06-06	22:00:00.440	Submission	Ask	\$4.300	16	3	1	5079468	NGN4

Note: This table presents an example of CME Market by Price (MBP) data for the Natural Gas futures contract (NGN4) on June 6, 2013. Each row represents a message describing an update to the limit order book (LOB). The messages include details about submissions, revisions, or deletions of orders at various price levels. In this example, all messages are categorized as "Submissions", indicating new entries into the LOB. The Date column specifies the date when the message was recorded, and the Time column provides the exact timestamp of the event expressed in milliseconds. The Type column indicates the nature of the market activity, such as submissions. The Side column identifies whether the event occurred on the bid or ask side of the market. The Price column shows the price level at which the activity took place, expressed in dollars per million British thermal units (MMBtu). The Size column specifies the number of contracts involved in each message. The Depth column indicates the relative price level within the LOB where the event occurred, with smaller depth values corresponding to price levels closer to the top of the book. The Order column represents the number of individual orders contributing to the total liquidity at the specified depth. The Message Seq # column provides a unique sequence number for each message, ensuring chronological tracking of market events. Finally, the Contract column identifies the futures contract using the base symbol NG for Natural Gas, N for the July contract, and 4 for the 2014 expiration. For instance, the first row describes a submission on the bid side at the 1st depth level with a price of \$4.102 /MMBtu, involving 1 contract and no individual orders. Similarly, the second row represents another bid-side submission at the 2nd depth level with a price of \$4.096 /MMBtu and 1 contract. The 5th row highlights a larger submission of 16 contracts on the bid side at the 2nd depth level for \$4.050 /MMBtu, contributed by 1 order.

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Appendix E

In this appendix we discuss the robustness of our parametric analysis. Due to the Quincy Data structural break, we split our sample into two periods: January 2011 to December 2012 and May 2013 to December 2017. We choose May 2013 as the start of the second period following Shkilko and Sokolov (2020) as Quincy Data began providing services around January 2013. This allows us to study the different HFT regimes separately; during the first period (2011-2012) few firms participated while in the second period (May 2013-2017), more firms likely employed spatial arbitrage trading.

Table E.1 offers regression results from equation (1) (described in the main text). It estimates the impact of BP(1 St.D) precipitation disruptions on market quality measures. Panel A reports results for the 2011–2012 period, and Panel B for the May 2013–2017 period. As discussed, the analysis uses minute-level data, incorporates fixed effects to control for security-specific and intraday patterns, and applies Driscoll-Kraay standard errors to account for cross-sectional dependence and autocorrelation (with L in parentheses indicating the number of lags used to model error term correlation). Lagged controls for volume and realized volatility are included to address potential endogeneity. Definitions and summaries of the market quality measures are provided in section 2.4 and Table 2 of the main text. Figure E.1 presents results for mean intraday realized volatility for our sample of ETFs. When performing this calculation we consider the full sample period 2011-2017. These inform our selection of dummy variables for regression equation (1) in the main text.

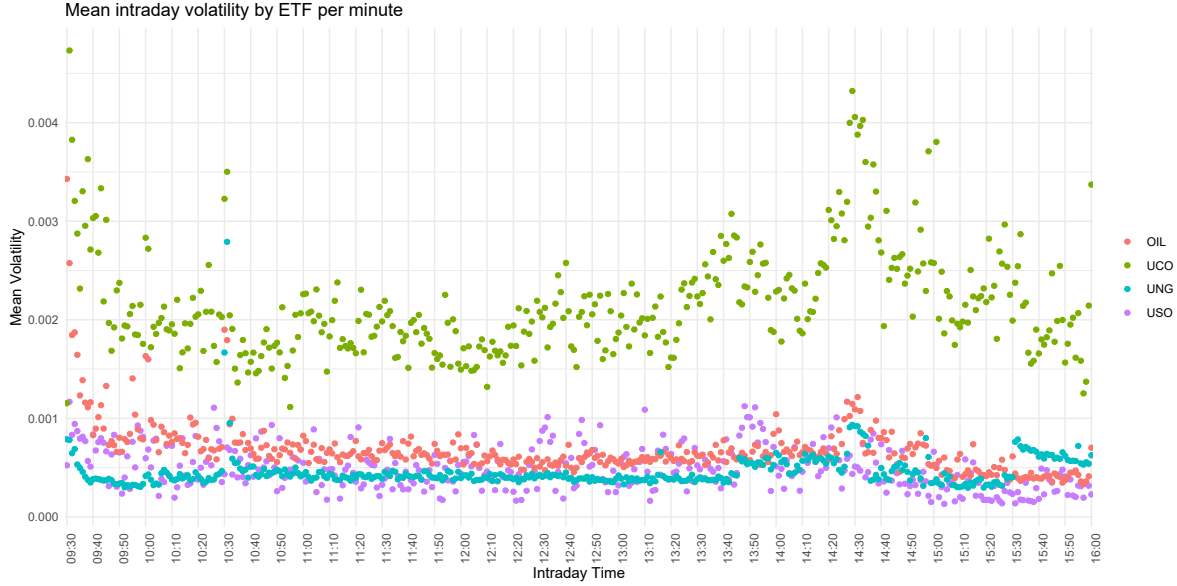


Fig. E.1. Mean intraday realized volatility by minute per security

Note: This figure presents results for mean intraday realized volatility for our sample of energy ETFs. When performing this calculation we consider the full sample period 2011-2017. Mathematical details for security realized volatility are provided in Table 3 and in section 3.4 of the main text. From the figure we can see that market open effects on volatility have dissipated between 9:40am and 9:50am while the closing effects begin around 5 minutes before market close at 3:55pm.

Table E.1. Regression results for BP(1 St.D)

Panel A: 2011-2012 USO-UNG	ES	RS 5s	PI 5s	RV	Ln(Bid MD)	Ln(Ask MD)	Ln(VOL)	NBBO	TS
BP(1 StD)	-0.0001* (5.46e-5)	-0.0001 (6.19e-5)	-1.6e-5 (3.52e-5)	0.0005*** (0.0001)	0.1931*** (0.0219)	0.1931*** (0.0219)	0.1524*** (0.0245)	6.836 (9.620)	1.127 (1.144)
Lag Ln(VOL)	-6.92e-5* (3.4e-5)	-0.0001 (5.41e-5)	5.76e-5 (3.34e-5)	5.21e-5* (2.29e-5)	0.1857*** (0.0045)	0.1857*** (0.0045)		128.9*** (2.713)	17.25*** (0.7408)
Lag RV	0.0154* (0.0077)	0.0144* (0.0056)	0.0019 (0.0041)		-3.137*** (0.5174)	-3.131*** (0.5174)	1.923** (0.5959)	595.2* (270.0)	321.2** (99.04)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)
Observations	155,294	155,294	155,294	155,294	155,294	155,294	155,317	155,294	155,294
R2	0.002	0.001	0.003	0.001	0.060	0.060	0.032	0.275	0.082
Adj. R2	0.002	0.001	0.003	0.001	0.060	0.060	0.032	0.275	0.082
F-statistic	19.1881	11.9031	31.3221	13.8876	725.3238	725.3280	430.4314	4261.6680	1000.2573
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Panel A: 2011-2012 UCO-OIL	ES	RS 5s	PI 5s	RV	Ln(Bid MD)	Ln(Ask MD)	Ln(VOL)	NBBO	TS
BP(1 StD)	-0.0004 (0.0008)	0.0004 (0.0003)	-0.0004 (0.0007)	0.0003 (0.0003)	0.0898*** (0.0228)	0.0896*** (0.0228)	0.1215*** (0.0252)	15.24 (10.96)	0.3898 (0.2710)
Lag Ln(VOL)	-0.0018*** (0.0004)	-0.0006*** (0.0002)	-0.0009** (0.0003)	6.13e-6 (5.76e-5)	0.1502*** (0.0051)	0.1502*** (0.0051)		53.20*** (2.239)	3.642*** (0.1125)
Lag RV	0.6181*** (0.2038)	0.0507*** (0.0194)	0.6388*** (0.2281)		-3.385*** (0.3726)	-3.393*** (0.3735)	-0.9055* (0.3723)	548.1*** (186.9)	-4.883 (4.142)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)
Observations	121,958	121,958	121,958	121,958	121,958	121,957	121,916	121,958	121,958
R2	0.008	0.010	0.008	0.011	0.130	0.131	0.115	0.134	0.123
Adj. R2	0.008	0.010	0.008	0.011	0.130	0.131	0.115	0.134	0.123
F-statistic	85.5139	104.6863	88.3762	143.4777	1588.6264	1593.6225	1611.5788	1637.4047	1488.1812
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Panel B: 2013-2017 USO-UNG	ES	RS 5s	PI 5s	RV	Ln(Bid MD)	Ln(Ask MD)	Ln(VOL)	NBBO	TS
BP(1 StD)	2.47e-6 (3.06e-5)	-5.56e-6 (3.19e-5)	7.06e-6 (1.04e-5)	8.65e-6 (7.82e-6)	-0.0470** (0.0200)	-0.0470** (0.0200)	-0.0133 (0.0216)	7.621 (4.326)	0.1456 (0.6424)
Lag Ln(VOL)	-4.26e-5*** (9.05e-6)	-6.52e-5*** (9.19e-6)	-1.1e-5*** (1.78e-6)	6.69e-6*** (1.94e-6)	0.1815*** (0.0020)	0.1815*** (0.0020)		61.08*** (0.8162)	13.07*** (0.1777)
Lag RV	0.6369*** (0.1830)	0.6209*** (0.1809)	0.0298* (0.0135)		-56.42*** (19.71)	-56.43*** (19.72)	15.50 (8.206)	5,898.8** (2,650.9)	1,212.2** (536.3)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=24)	DK (L=24)	DK (L=24)	DK (L=24)	DK (L=24)	DK (L=24)	DK (L=24)	DK (L=24)	DK (L=24)
Observations	637,259	637,259	637,259	637,259	637,259	637,258	637,400	637,259	637,259
R2	0.050	0.037	0.001	0.003	0.241	0.241	0.071	0.196	0.140
Adj. R2	0.050	0.037	0.001	0.003	0.241	0.241	0.071	0.196	0.140
F-statistic	2480.4865	1831.9747	52.2698	163.9420	15095.8865	15096.5138	4269.8374	11589.1560	7749.7037
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Panel B: 2013-2017 UCO-OIL	ES	RS 5s	PI 5s	RV	Ln(Bid MD)	Ln(Ask MD)	Ln(VOL)	NBBO	TS
BP(1 StD)	8.45e-5 (0.0001)	1.65e-5 (8.38e-5)	7.09e-5 (7.66e-5)	-6.82e-5*** (1.84e-5)	-0.0162 (0.0266)	-0.0170 (0.0267)	-0.0339 (0.0270)	8.993 (5.003)	-0.5614 (0.3767)
Lag Ln(VOL)	-0.0007*** (2.27e-5)	-0.0006*** (1.71e-5)	-0.0002*** (1.44e-5)	-6.64e-6 (4.31e-6)	0.2315*** (0.0029)	0.2307*** (0.0029)		58.03*** (0.8815)	6.703*** (0.0879)
Lag RV	0.8520*** (0.2756)	0.4272*** (0.1268)	0.1537 (0.0788)		-31.34*** (3.995)	-31.68*** (4.045)	-2.342* (1.014)	385.5 (303.7)	78.94** (31.69)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=23)	DK (L=23)	DK (L=23)	DK (L=23)	DK (L=23)	DK (L=23)	DK (L=23)	DK (L=23)	DK (L=23)
Observations	546,095	546,095	546,095	546,095	546,095	546,082	546,279	546,095	546,095
R2	0.023	0.029	0.001	0.001	0.085	0.085	0.091	0.233	0.137
Adj. R2	0.023	0.029	0.001	0.001	0.085	0.085	0.091	0.233	0.137
F-statistic	1037.0699	1326.5316	57.0992	68.9821	4168.5565	4142.5343	5249.5521	13661.3244	7116.6784
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Note: This table presents panel regression results from equation (1) (described in the main text), estimating the impact of BP(1 St.D) precipitation disruptions on market quality measures. The analysis uses minute-level data, incorporates fixed effects to control for security-specific and intraday patterns, and applies Driscoll-Kraay standard errors to account for cross-sectional dependence and autocorrelation (with L in parentheses indicating the number of lags used to model error term correlation). Lagged controls for volume and realized volatility are included to address potential endogeneity.

Table E.2. Robustness Checks for Regression results for BP(ITU)*

Panel A: 2011–2012 USO–UNG Horizon Robustness	RS 10s	PI 10s	RS 15s	PI 15s	RS 60s	PI 60s
BP(ITU)*	-2.38e-5 (0.0001)	1.27e-5 (9.5e-5)	6e-5 (0.0001)	-7.24e-5 (0.0001)	-0.0002 (0.0004)	0.0001 (0.0004)
Lag Ln(VOL)	-1.73e-5 (1.81e-5)	-2.03e-6 (1.91e-5)	-2.42e-5 (2.18e-5)	4.46e-6 (2.31e-5)	-3.61e-5 (5.34e-5)	1.51e-5 (5.46e-5)
Lag RV	0.0010 (0.0022)	0.0041 (0.0023)	0.0032 (0.0030)	0.0014 (0.0032)	0.0011 (0.0077)	0.0045 (0.0082)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)
Observations	148,374	148,374	148,374	148,374	148,374	148,374
R2	0.0003	0.003	0.0004	0.001	0.0010	0.0002
Adj. R2	0.0003	0.003	0.0003	0.001	0.0009	0.0002
F-statistic	3.6626	34.9561	4.1395	14.0343	10.6146	2.1596
F p-value	0.0006	0.0000	0.0001	0.0000	0.0000	0.0345

Panel A: 2011–2012 UCO–OIL Horizon Robustness	RS 10s	PI 10s	RS 15s	PI 15s	RS 60s	PI 60s
BP(ITU)*	6.66e-5 (0.0006)	-0.0016** (0.0005)	0.0004 (0.0006)	-0.0012 (0.0011)	6.25e-5 (0.0013)	-0.0021 (0.0012)
Lag Ln(VOL)	-0.0006*** (0.0002)	-0.0013*** (0.0005)	-0.0006*** (0.0002)	-0.0015*** (0.0005)	-0.0004** (0.0002)	-0.0016*** (0.0005)
Lag RV	0.0510*** (0.0180)	0.2740** (0.1148)	0.0510*** (0.0185)	0.4550*** (0.1515)	0.0558** (0.0235)	0.3725*** (0.1359)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)	DK (L=16)
Observations	119,184	119,184	119,184	119,184	119,184	119,184
R2	0.012	0.002	0.008	0.004	0.005	0.003
Adj. R2	0.012	0.002	0.008	0.004	0.005	0.003
F-statistic	122.4131	24.0275	78.5502	43.6298	54.7343	26.4045
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Panel B: 2013–2017 USO–UNG Horizon Robustness	RS 10s	PI 10s	RS 15s	PI 15s	RS 60s	PI 60s
BP(ITU)*	-5.79e-5 (4.63e-5)	8.72e-6 (3.1e-5)	-5.74e-5 (5.31e-5)	1.23e-5 (4.13e-5)	-4.53e-5 (0.0001)	-3.17e-5 (0.0001)
Lag Ln(VOL)	-0.0001*** (1.06e-5)	-6.27e-7 (3.72e-6)	-0.0001*** (1.12e-5)	2.57e-5*** (5.19e-6)	-0.0002*** (1.88e-5)	8.7e-5*** (1.9e-5)
Lag RV	0.5457*** (0.1645)	0.0430* (0.0206)	0.5468*** (0.1644)	0.0463* (0.0223)	0.3637 (0.2105)	0.2942 (0.1517)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)
Observations	513,324	513,324	513,324	513,324	513,324	513,324
R2	0.022	0.001	0.016	0.0004	0.002	0.001
Adj. R2	0.022	0.0005	0.016	0.0004	0.002	0.0008
F-statistic	860.4577	19.9743	617.0941	15.4249	60.6273	29.2307
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Panel B: 2013–2017 UCO–OIL Horizon Robustness	RS 10s	PI 10s	RS 15s	PI 15s	RS 60s	PI 60s
BP(ITU)*	-0.0003** (0.0001)	-3.8e-5 (8.09e-5)	-0.0003** (0.0001)	-3.8e-5 (9.53e-5)	-0.0005** (0.0002)	0.0002 (0.0002)
Lag Ln(VOL)	-0.0006*** (2.03e-5)	-0.0002*** (1.31e-5)	-0.0005*** (2.13e-5)	-0.0003*** (1.67e-5)	-0.0005*** (2.88e-5)	-0.0003*** (3.02e-5)
Lag RV	0.4385*** (0.1472)	0.1778 (0.1052)	0.4293*** (0.1469)	0.1874 (0.1064)	0.7327** (0.3551)	0.7166 (0.3730)
F.E. Ticker	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Open	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Midday	Yes	Yes	Yes	Yes	Yes	Yes
F.E. Market Close	Yes	Yes	Yes	Yes	Yes	Yes
S.E. Type	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)	DK (L=22)
Observations	436,204	436,204	436,204	436,204	436,204	436,204
R2	0.019	0.003	0.012	0.002	0.008	0.005
Adj. R2	0.019	0.003	0.012	0.002	0.008	0.005
F-statistic	705.2901	107.3693	457.4382	66.9005	279.4219	193.8068
F p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Note: This table presents horizon robustness for increasing δ in RS and PI panel regression results from equation (1) (described in the main text), estimating the impact of BP(ITU)* precipitation disruptions on market quality measures. The analysis uses minute-level data, incorporates fixed effects to control for security-specific and intraday patterns, and applies Driscoll-Kraay standard errors to account for cross-sectional dependence and autocorrelation (with L in parentheses indicating the number of lags used to model error term correlation). Lagged controls for volume and realized volatility are included to address potential endogeneity.

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